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NeuroAudit

AI Secure Code Reviewer

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Abstract

Security weaknesses in web applications frequently stem from vulnerabilities in PHP code, particularly related to SQL Injection (SQLi) and Cross-Site Scripting (XSS). As development practices increasingly incorporate Continuous Integration and Continuous Deployment (CI/CD), there has been a notable rise in the necessity for automated tools that can identify these issues. These tools not only enhance application security but also minimize the substantial time and effort traditionally spent on manual code inspections, which are typically resource-intensive and costly. This document introduces NeuroAudit, a proficient framework driven by neural networks designed to independently evaluate PHP source code to uncover potential SQL Injection (SQLi) and Cross-Site Scripting (XSS) vulnerabilities. The model performs multi-class classification to determine if a segment of code is secure or vulnerable, and if identified as vulnerable, it identifies the specific type of vulnerability. The detection begins with converting the code into an Abstract Syntax Tree (AST), which represents both the syntactic structure and semantic elements of the program. This representation is then tokenized and analyzed using a neural network system that combines Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (BiLSTM) networks. This configuration preserves the code sequence while effectively capturing long-distance dependencies. We evaluated the performance of the system using the PHP Test Suite – XSS, SQLi 1.0.0 (2022) supplied by the Software Assurance Reference Database (SARD). NeuroAudit demonstrated impressive performance, achieving precision, recall, and F1-scores of 82%, 98%, and 90% for SQLi; 100%, 97%, and 99% for XSS; and 96% across all metrics for multi-class classification. Additionally, we confirmed the efficacy of the model using real-world examples sourced from GitHub repositories and Exploit DB, thereby validating its dependability and effectiveness in real-world applications.

Attestation &Turnitin Report

I understand the nature of plagiarism, and I am aware of the University's policy on this.

I certify that this dissertation reports original work by me during my University project except for the following (*adjust according to the circumstances*):

- The technology review in Section 2.5 was largely taken from [17].
- The code discussed in Section 3.1 was created by Acme Corporation (*www.acme-corp.com/JavaExpert*) and was used in accordance with the licence supplied.
- The code discussed in Section 3.5 was written by my supervisor.
- The code discussed in Section 4.2 was developed by me during a vacation placement with the collaborating company. In addition, this used ideas I had already developed in my own time.

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1 Chapter 1 - Introduction

1.1 Overview

In recent years, the growing reliance on web applications has brought software security to the forefront of development priorities. Despite the increasing awareness of secure coding practices, many applications remain vulnerable to common attacks such as SQL Injection (SQLi) and Cross-Site Scripting (XSS), particularly in dynamic languages like PHP. These vulnerabilities can be exploited to compromise user data, manipulate databases, or gain unauthorized access to systems. Traditional code review methods are often labor-intensive, subjective, and may fail to detect hidden or complex vulnerabilities.

To address this challenge, our graduation project introduces **NeuroAudit – AI Secure Code Reviewer**, a deep learning–based solution designed to automate the detection of SQLi and XSS vulnerabilities in PHP source code. By utilizing advanced models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM), the system is capable of learning contextual and structural patterns within the code to effectively identify security threats. NeuroAudit aims to improve accuracy in secure code auditing while reducing manual effort and human error.

1.2 Problem Statement

PHP is a widely used language in web development, yet it is often criticized for being prone to security vulnerabilities due to its flexible syntax and dynamic features. Many applications built in PHP contain vulnerabilities that may not be immediately apparent, especially in large codebases. Manual detection methods are not only time-consuming but also susceptible to human error. The absence of efficient and automated tools for vulnerability detection increases the likelihood of exploitable code reaching production environments.

1.3 Scope and Objectives

The scope of this project focuses on analyzing PHP source code to identify two major types of vulnerabilities: SQL Injection and Cross-Site Scripting. The primary objectives of the project are as follows:

- To build and train a deep learning model using CNN and LSTM architectures to classify PHP scripts as vulnerable or secure.
- To develop a system that automatically detects the presence of SQLi and XSS vulnerabilities in raw PHP code files.

- To design a user-friendly interface that facilitates the secure code review process.
- To evaluate the model's performance in terms of accuracy and reliability using real and synthetic data samples.

The project is limited to static analysis of PHP code and does not include dynamic testing, patch recommendation, or vulnerability exploitation simulation.

1.4 Report Organization (Structure)

This report is organized into seven main chapters. Chapter 1 provides an introduction to the project, including the background, problem statement, objectives, and methodology. Chapter 2 reviews related work and existing literature in the field of vulnerability detection and machine learning in security. Chapter 3 outlines the proposed solution, including the system design and functional requirements. Chapter 4 covers implementation details, followed by Chapter 5, which presents the testing approach and evaluation metrics. Chapter 6 discusses the results and their significance. Finally, Chapter 7 summarizes the project and explores potential areas for future development.

Briefly overview the contents of what follows in the dissertation.

1.5 Work Methodology

Our approach to solving the problem follows a structured methodology involving several key stages:

- **Research and Data Collection**: Gathering PHP code samples, both vulnerable and secure, from public repositories, vulnerability databases, and academic sources.
- **Data Preprocessing**: Tokenizing and transforming the code into formats suitable for deep learning input, while preserving code structure and semantic relevance.
- **Model Development**: Designing and training a hybrid deep learning model combining Convolutional Neural Networks (CNN) for pattern extraction and Long Short-Term Memory (LSTM) networks for sequence modeling. This approach enables the system to capture both spatial and contextual information from the code.
- **System Integration**: Developing a backend system that integrates the trained model, allowing for real-time analysis of uploaded PHP files.
- **Testing and Evaluation**: Measuring the system's performance using various metrics such as accuracy, precision, recall, and F1-score across test datasets.

- **Documentation and Reporting**: Compiling findings, results, design decisions, and insights into a structured graduation project report.

2 Chapter 2 - Related Work (State-of-The-Art)

This section elaborates on key ideas related to identifying vulnerabilities, covering Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Abstract Syntax Tree (AST) representation. The section wraps up by presenting a synopsis of significant academic publications.

2.1 Background

Deep Neural Network (DNN):
This refers to a category of machine learning models characterized by multiple layers of interconnected nodes or neurons, where each layer adjusts its input data through learned weights and nonlinear activation functions. The word 'deep' refers to the multiple hidden layers situated between the input and output layers, which improve the network's capacity to learn complex, hierarchical data representations.

Key Characteristics:

- Hierarchical Feature Learning: Deep neural networks systematically extract more abstract features from raw data as it moves through layers, thus boosting their effectiveness.
- Nonlinear Transformations: Each neuron applies a nonlinear activation function, enabling the network to capture complex, nonlinear relationships in the data.
- End-to-End Learning: Deep Neural Networks are trained on large datasets and a variety of optimization methods, including backpropagation, to improve weights and minimize prediction errors.

Convolutional Neural Network (CNN):
This model is a specialized form of deep learning designed to automatically learn hierarchical features from unprocessed data, especially images. CNNs include several layers, such as convolutional layers which use filters to discover local patterns, pooling layers that reduce spatial dimensions, and fully connected layers that handle final classification tasks. This architecture enables CNNs to independently learn complex data representations, eliminating the necessity for manual feature engineering.

Principal Characteristics:

- Automatic Feature Extraction: Convolutional Neural Networks (CNNs) succeed in capturing features that range from simple to complex.
- Efficiency and Robustness: Convolutional Neural Networks (CNNs) are adept at processing large datasets efficiently and show resilience to variations in input, including noise or changes in the background.

Recurrent Neural Network (RNN):
This type of neural network is designed to evaluate sequential data by maintaining a memory of previous inputs through recurrent connections. This structure enables RNNs to grasp temporal relationships and context within sequences, making them particularly useful for tasks where the sequence order matters. RNNs implement weight-sharing across time steps, enabling efficient learning of patterns in sequential data and retaining information from earlier elements in the sequence for later predictions.

Key Characteristics:

- Sequential Memory: Recurrent Neural Networks (RNNs) maintain a memory function that allows them to use historical context for future predictions.
- Weight Sharing: The cyclic structure of the network supports effective learning and generalization across different points within the sequence.
- Limitations: Traditional recurrent neural networks encounter difficulties in retaining information over long sequences due to issues like vanishing gradients, which can limit their ability to model long-term dependencies.

Long Short-Term Memory (LSTM):
LSTM is a sophisticated version of recurrent neural networks (RNNs) created to address the shortcomings of traditional RNNs when it comes to learning long-term connections in sequential information. It incorporates gating mechanisms into its cell design, allowing it to effectively hold onto and utilize crucial data over extended input sequences, a capability that traditional RNNs using basic activation functions lack. The design of an LSTM cell consists of input, output, and forget gates along with state memory, which improves the way it models data based on both time and sequence.

Key features:
- Memory cells: LSTM incorporates specialized units known as memory cells that can retain information over long durations.
- Gating mechanisms: Long Short-Term Memory networks feature input gates, output gates, and forget gates that manage the flow of information in and out of the memory cells.
- Constant error flow: LSTM makes use of constant error carousels (CECs) to maintain a steady error flow within specific cells, thus aiding in connecting long time gaps.
- Ability to learn long-term dependencies: LSTM showcases the ability to learn how to navigate minimal time intervals exceeding 1,000 distinct time steps.
- Forget gates: LSTM features forget gates that can learn to reset the internal state of memory cells when the stored information is no longer relevant.

Abstract Syntax Tree (AST) Representation:
An Abstract Syntax Tree (AST) is a well-organized, tree-like structure that depicts the syntactic framework of source code, showcasing how its various elements relate to one another. ASTs play a vital role in code representation learning, as they encapsulate both the syntax and, to some extent, the semantics of programs, making them widely used in activities like program analysis, classification, and defect prediction.

Key characteristics:
- Hierarchical Organization: Nodes represent elements such as statements, expressions, and control structures.
- Rich Feature Encoding: Abstract Syntax Trees (ASTs) capture both basic and complex features.
- Semantic Information: By illustrating the code structure and relationships, Abstract Syntax Trees (ASTs) assist in preserving semantic information.

2.2 Literature Survey

A broad spectrum of approaches in recent literature addresses vulnerability detection:

- **Graph Neural Networks and Token-Based Methods:**

  VulEye: A Novel Graph Neural Network Vulnerability Detection Approach for PHP Application

and works like *RecurScan: Detecting Recurring Vulnerabilities in PHP Web Applications* explore graph-based representations of source code to capture its structural properties. Additionally, *A Static Analysis Model for PHP Vulnerabilities Based on Token and Deep Learning Technology* uses tokenization paired with deep learning to focus on language-specific patterns for PHP.

- **Deep Learning Models (CNN, LSTM, and Hybrid Approaches):**
  Several studies leverage deep architectures to automatically learn features from code. For instance, *VulHunter: An Automated Vulnerability Detection System Based on Deep Learning and Bytecode* and *VulDeePecker: A Deep Learning-Based System for Vulnerability Detection* demonstrates the efficacy of CNNs and LSTMs in identifying vulnerabilities. The hybrid model in *DB-CBIL: A DistilBert-Based Transformer Hybrid Model Using CNN and BiLSTM for Software Vulnerability Detection* exemplifies the trend toward combining spatial and sequential modeling for enhanced performance.

- **NLP and Machine Learning Techniques:**
  Approaches such as *Comparison Study for NLP Using Machine Learning Techniques to Detect SQL Injection Vulnerabilities* focus on applying Natural Language Processing (NLP) methods to analyze code patterns. Other works like *Vulnerability Detection Under Poisoning Attacks Through Code and Token Features* and *Detecting and Removing Web Application Vulnerabilities with Static Analysis and Data Mining* incorporate traditional machine learning methodologies alongside deep learning frameworks.

- **Innovative and Emerging Methods:**
  The literature also covers emerging themes such as vulnerability prediction using software metrics (e.g., *Vulnerability Prediction From Source Code Using Machine Learning*), code similarity analysis (*VulPecker: An Automated Vulnerability Detection System Based on Code Similarity Analysis*), and even integrated approaches for detecting and repairing vulnerabilities—as highlighted by recent efforts like *ChatGPT for Vulnerability Detection, Classification, and Repair: How Far Are We?* and *Large Language Model for Vulnerability Detection and Repair: Literature Review and the Road Ahead*.

Other notable works include *DeepTective*, *DEKANT: A Static Analysis Tool that Learns to Detect Web Application*, *Context-Aware Software Vulnerability Classification Using Machine Learning*, *CODDLE: Code-Injection Detection with Deep Learning*, *Vulnerable Code Detection Using Software Metrics and Machine Learning*, and several others that expand on these themes.

2.3 Analysis of the Related Work

Static Analysis Tools and Vulnerability Scanners:

- Static source-code analysis has long been applied to PHP vulnerability detection. Tools like RIPS perform tokenization and taint analysis on PHP ASTs to detect sensitive sinks (e.g. SQL queries, echo statements) vulnerable to user input. Earlier tools such as Pixy and PHP-SAT also used lightweight taint analysis (often regex-based) to flag XSS or SQLi patterns, but they handled only single files or simple code, leading to many false positive. Many open-source PHP scanners exist: for example, Progpilot (2018, MIT license) is a static taint analyzer covering SQLi and XSS with configurable sinks and sanitizers, and tools like Semgrep (with OWASP rules) allow custom pattern matching Industry platforms like SonarQube and GitLab SAST include PHP vulnerability rules (OWASP Top 10, CWE checks), and commercial scanners (e.g. Checkmarx, Veracode) support PHP as well. While such static tools are widely used in practice, published evaluations on PHP are limited; they are known to generate many false positives and require expert tuning to reduce noise.

Dynamic Analysis and Heuristic Methods:

- In parallel, dynamic or black box testing methods are used to find SQLi/XSS in web apps. Security scanners and fuzzers (e.g. SQLMap, OWASP ZAP, Arachni) inject attack payloads against running PHP applications to discover exploitable inputs. These approaches can confirm real vulnerabilities but require a live environment and cannot directly label source code. Heuristic or pattern-based techniques (e.g. PHP lint rules, custom grep checks, web application firewalls) look for common insecure code idioms (like mysql_query($_GET […) without sanitization). Such methods are simple to implement but have limited coverage and often miss context-dependent flaws. In summary, traditional heuristics can raise obvious alarms quickly but lack the sophisticated data-flow reasoning of full static analysis.

Machine Learning and Deep Learning Approaches:

- Recent research has applied machine learning (ML) and AI to PHP vulnerability classification. Early learning-based models treated vulnerability detection as a classification problem. For example, DEKANT (Medeiros et al., 2016) used a Hidden Markov Model on sequences of code tokens (marked tainted/non-tainted) to characterize PHP vulnerabilities. Similarly, the WAP/WEAPONS

framework applied taint analysis with logistic regression and other ML classifiers across multiple vulnerability types, yielding high recall on synthetic samples. Building on these, WIRECAML (Kronjee et al., 2018) extracted rich data-flow features (reaching definitions, taints, constants) from PHP code graphs and trained decision trees. On standard test suites, WIRECAML outperformed static tools (RIPS, Pixy) on most metrics (precision, recall, F1). These data-driven models demonstrate that combining static flow analysis with ML can reduce manual rule-crafting and catch complex patterns better than pure heuristics.

- More recent work uses deep learning to automatically learn code representations. TAP (Fang et al., 2019) tokenizes PHP and performs simple dataflow slicing, then embeds tokens with word2vec and classifies with an LSTM network, substantially outperforming RIPS in multiclass detection. Likewise, VulHunter (Ahmad et al., 2020) generates CFG/DFG slices around sensitive calls, encodes them (token + bytecode features) and uses a Bi-LSTM; it reports higher recall and F1 on SQLi/XSS than RIP, even discovering new vulnerabilities in real PHP CMS projects. These deep models remove the need for hand-crafted rules and can capture broader contextual patterns, though they typically require large, labeled corpora.
- The latest trend is to use graph neural networks (GNNs) on code graphs. DeepTective (Rabheru 2020) was the first to apply a hybrid GNN (combining GRU and graph convolutions) to PHP; it models the program CFG to predict XSS/SQLi but has limited public metrics. Most notably, VulEye (Lin et al., 2022) constructs a sub-dependence graph (combining control/data dependencies plus token embeddings) around each sink and applies a graph convolutional network with attention. Overall, AI-driven models (RNNs, GNNs) have demonstrated superior accuracy on benchmark datasets compared to traditional static tools, but they often rely on synthetic or project-specific data and can struggle to generalize to entirely new codebases.

3 Chapter 3 - Proposed solution

This chapter outlines the technical design, methodology, and architecture of *NeuroAudit – AI Secure Code Reviewer*, a deep learning-based system developed to detect vulnerabilities in PHP source code. The project focuses on two high-impact vulnerabilities: **SQL Injection (SQLi)** and **Cross-Site Scripting (XSS)**. The solution comprises three specialized neural network models trained to classify source code as secure or vulnerable.

The motivation behind the proposed approach is to improve the automation and precision of static code analysis using advanced deep learning techniques such as **Convolutional Neural Networks (CNNs)** and **Long Short-Term Memory (LSTM)** networks. By combining spatial and sequential pattern recognition, the models can better capture both syntactic structures and contextual flow within PHP programs.

3.1 Solution Methodology

To tackle the challenge of automated vulnerability detection, *NeuroAudit* uses a supervised learning strategy built upon a combination of public and custom PHP vulnerability datasets. The methodology follows these key stages:

3.1.1 Data Sources

To ensure robustness and generalizability, we utilized a combination of publicly available and custom-built datasets. These include large-scale vulnerability corpora, fine-tuned filtered subsets, and out-of-distribution test sets, as outlined below:

PHP Vulnerability Test Suite (2015)

- **Overview & Purpose:**

This dataset is part of NIST's SARD (Software Assurance Reference Dataset) and in-cludes approximately 42,000 synthetic PHP samples. It was designed to cover a wide spectrum of vulnerability types, including XSS, SQL Injection, URL redirection, and more.

- **Strengths:**

Broad coverage across ~10 CWE categories.

High diversity due to generator-based creation by Telecom Nancy (Stivalet & Delaitre, ICST 2016).

- **Challenges:**

Requires filtering and cleaning to extract SQLi and XSS samples.

May not reflect complex real-world programming patterns.

- Source:

https://samate.nist.gov/SARD/test-suites/103

PHP Test Suite – XSS, SQLi 1.0.0 (2022)

- **Overview & Purpose**:
  Also from NIST's SARD, this more recent dataset consists of ~**248,600 PHP files** designed specifically for detecting **SQL Injection and XSS vulnerabilities**.
- **Strengths**:
  - Directly focused on the two target vulnerability classes (SQLi, XSS).
  - Consistent structure and annotation suitable for deep learning.
- **Size**:
  ~931 MB (archive), with SHA-256 checksum provided.
- **Suitability**:
  Ideal for training and validating models due to its scale and specificity.
- **Sources**:
  - NIST: https://samate.nist.gov/SARD/test-suites/114
  - Related publication: https://yuanxzhang.github.io/paper/recurscan-www24.pdf

Custom Out-of-Sample Evaluation Sets

- To evaluate generalization beyond training data and simulate real-world scenarios, we constructed two additional datasets:
- **Dataset A**:
  - **Composition**: 140 manually curated PHP samples, balanced between secure and SQLi-vulnerable snippets.

    - **Purpose**: Used for final testing of the binary SQLi model on realistic, out-of-distribution samples.

- **Dataset B**:
    - **Composition**: 69 real SQLi-vulnerable examples and 69 hand-sanitized, secure counterparts.
    - **Purpose**: Used to test model sensitivity to input sanitization techniques and confirm behavioral distinctions in predictions.
- These datasets serve as stress tests to evaluate the robustness of both binary and multi-class classifiers outside controlled training environments.

3.1.2 Preprocessing Pipeline

To prepare the PHP code for input into deep learning models, a structured preprocessing pipeline was developed. The objective was to preserve both the **syntactic integrity** and the **semantic meaning** of each code snippet while transforming it into a machine-readable format. This pipeline was consistently applied across all datasets used in training, validation, and evaluation.

The preprocessing process involved four main stages:

Tokenization

- Each PHP file was first broken down into a sequence of atomic components. These **tokens** included:
- **Keywords** (e.g., if, while, echo)
- **Operators** (e.g., =, +, ==)
- **Identifiers** (e.g., variable names, function names)
- **Constants and symbols** (e.g., string literals, semicolons)
- This step normalized code syntax, reduced semantic noise, and allowed the models to focus on the structure and patterns relevant to vulnerability detection.

Abstract Syntax Tree (AST) Parsing

- To extract deeper structural and semantic information, the code was optionally parsed into **Abstract Syntax Trees (ASTs)**—particularly in the **binary SQLi classifier**.
- The ASTs were **linearized** into ordered sequences of node types and labels, capturing the hierarchical and logical flow of the program.
- This allowed the model to learn relationships that go beyond simple token order, including **control flow**, **data flow**, and **nesting**.
- This representation proved especially useful for detecting vulnerabilities that depend on how data propagates through the code (e.g., user input into SQL queries).

Sequence Padding

- To ensure uniform input dimensions during batch processing, all tokenized (or AST-linearized) sequences were padded to a **max length of all inputs**:
- Padding was applied **post-sequence** to preserve early-context tokens, which are often semantically significant (e.g., variable declarations, input capture).
- This consistency allowed all input vectors to be batched efficiently without compromising contextual learning.

Numerical Encoding and Embedding

- Once standardized, tokens and AST nodes were transformed into **integer indices** using a vocabulary dictionary built from the dataset.
- These integers were then passed through a **learnable embedding layer**, producing dense vector representations of size 128.
- Embedding allowed the model to learn **semantic similarity** between tokens during training (e.g., different input variable names that behave similarly across examples).

Impact of Preprocessing on Model Capability

- This layered preprocessing strategy allowed our deep learning models to:
- Recognize **surface-level patterns** and syntax via CNNs (e.g., typical SQL query structures)

- Learn **long-range dependencies** and control flow through LSTMs (e.g., how user input is processed and passed)
- Capture **abstract program logic and structure** via ASTs (e.g., data propagation, branching, nesting)
- As a result, *NeuroAudit* achieved high accuracy in distinguishing between secure and vulnerable PHP code—even in obfuscated or unconventional scripts.

3.1.3 Model Architecture

All three *NeuroAudit* models share a **CNN–BiLSTM hybrid architecture** tailored to AST sequence inputs. This allows the system to detect both **local syntax anomalies** and **global vulnerability patterns**.

Model Types:

- **Model 1 – SQLi Classifier:**

  A binary classifier that determines whether a PHP snippet contains a SQL Injection vulnerability.

- **Model 2 – XSS Classifier:**

  A binary classifier dedicated to detecting Cross-Site Scripting (XSS) vulnerabilities.

- **Model 3 – Multi-class Classifier:**

  A multi-class classifier that identifies if a code snippet is Safe, SQLi-vulnerable, or XSS-vulnerable.

All three models follow a common hybrid architecture combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) layers. This design effectively captures both spatial patterns in the code structure and sequential dependencies across tokens or AST nodes.

Architecture Overview

Layer	Description
Input	Linearized AST node sequence
Embedding	Embedding (input_dim=V, output_dim=128, trainable=True) – Converts token indices to dense vectors
Conv1D	Conv1D (filters=64, kernel_size=3, activation='relu', padding='same') – Extracts local features
BiLSTM	Bidirectional (LSTM (32, return_sequences=True, kernel_regularizer=l1_l2(1e-6, 1e-6)) – Models contextual flow in both directions
GlobalMax-Pooling1D	Aggregates strongest signal across all time steps, preserving key features
Dropout Layer	A dropout rate of 0.5 is applied after the dense layer to prevent overfitting by randomly disabling 50% of the neurons during training.
Dense Layer	Dense (32) – Intermediate fully connected layer
Output Layer	- Binary: Dense (1, activation='sigmoid') - Multi-class: Dense (3, activation='softmax')

Design Highlights

- **AST-Based Inputs**: All models use syntactic trees instead of raw tokens, improving structure comprehension.
- **Convolutional Layer**: Detects localized patterns like unsanitized input or risky API calls.
- **Bidirectional LSTM**: Captures forward and backward code dependencies, enhancing logic flow modeling.

- **GlobalMaxPooling**: Retains salient features across the sequence, mitigating the vanishing gradient issue.
- **Dropout Layer**: A dropout to prevent overfitting by randomly disabling 50% of the neurons during training.
- **Regularization**: l1_l2 regularization in LSTM combats overfitting with sparsity and stability.

Training Configuration

- **Loss Functions**:
    - Binary models: binary_crossentropy
    - Multi-class model: categorical_crossentropy
- **Optimizer**: Adam(learning_rate=0.01)
- **Epochs**: Up to 4 with EarlyStopping on validation F1-score
- **Batch Size**: 128

Evaluation Metrics

All models are evaluated using:

- Accuracy
- F1 Score (Macro-average for multi-class)
- Precision
- Recall
- Loss

3.2 Functional / Non-functional Requirements

Functional Requirements

- **The system shall accept raw PHP files or code snippets as input.**
- **It shall detect SQLi and XSS vulnerabilities using deep learning models.**
- **It shall support two analysis modes:**

- **Binary detection (SQLi or XSS)**
- **Multi-class detection (Secure, SQLi, or XSS)**

- **It shall present results in a clear and interpretable format to users.**

Non-Functional Requirements

- **Performance: Results should be returned within 2 seconds for average-sized files.**
- **Accuracy: The system must demonstrate high precision and recall on evaluation datasets.**
- **Security: User-submitted code must be handled securely and deleted after processing.**
- **Usability: The interface must be simple and accessible to non-technical users.**
- **Maintainability: The architecture should allow for future extension (e.g., more vulnerability types).**
- **Portability: The system should be deployable on both local and cloud environments.**

3.3 Design / Simulation set up

3.3.1 System Architecture

The system is divided into the following components:

- **Frontend Interface**: A user-friendly web application for uploading and analyzing PHP files.
- **Backend API**: Built using Flask, it processes incoming requests, applies preprocessing, and routes data to the appropriate model.
- **Model Management Unit**: Selects between SQLi binary, XSS binary, or multi-class models based on analysis type.
- **Deep Learning Core**: Implements CNN-LSTM hybrid models to classify code.
- **Response Module**: Converts model outputs into readable feedback (e.g., “Secure”, “Vulnerable: SQLi”).

3.3.2 Model Architecture

Each of the three models shares the following architecture:

- **Embedding Layer:**

  Maps input tokens to 128-dimensional dense vectors, allowing the model to learn semantic representations of code tokens.

- **CNN Layer:**

  Applies 64 convolutional filters with a kernel size of 3 to capture local patterns and vulnerability-specific token sequences in the input.

- **Bidirectional LSTM (BLSTM) Layer:**

  32 hidden units in each direction to capture long-range dependencies and the contextual flow of code both forward and backward.

- **Global Max Pooling Layer:**

  Reduces the dimensionality of the output sequence by selecting the maximum value from each feature map across the entire sequence, helping highlight the most important features and reduce overfitting.

- **Dropout Layer:**

  A dropout rate of 0.5 is applied after the dense layer to prevent overfitting by randomly disabling 50% of the neurons during training.

- **Fully Connected (Dense) Layer**:

  A dense layer with 32 neurons to further process the pooled features before classification.

- **Output Layer:**

  Binary classifiers use a single neuron with sigmoid activation to output a probability indicating whether the code is safe or vulnerable.

Multi-class classifier uses a softmax activation with 3 output neurons, corresponding to the classes: Safe, SQL Injection, and XSS.

3.3.3 Training Configuration

Language: Python 3.10

Libraries:

- TensorFlow: 2.13
- Keras API (within TensorFlow)
- Scikit-learn
- Gensim (for Word2Vec – not shown in full code yet, but imported)
- Matplotlib & Seaborn (for visualization)

Optimizer: Adam

Learning Rate: 0.01

Loss Function:

- Binary cross-entropy (binary models)
- Categorical cross-entropy (multi-class)

Batch Size: 128

Epochs: 4

Validation Split: 13%

Regularization: Early stopping

Hardware Environment

Platform: Kaggle Notebook (GPU-accelerated)

GPU: NVIDIA Tesla P100 (16 GB VRAM)

RAM: 29 GB

Storage: 57.6 GB

Evaluation Metrics(weighted)

SQLi Binary Model:

- Accuracy: 95.3%
- F1-Score: 95.0%
- Precision: 96.0%
- Recall: 95.0%
- Test loss:0.11

XSS Binary Model:

- Accuracy: 98.8%
- F1-Score: 99.0%
- Precision: 99.0%
- Recall: 99.0%
- Test loss:0.02

Multi-Class Model:

- Accuracy: 97.0%
- F1-Score: 97.0%
- Precision: 98.0%
- Recall: 97.0%
- Test loss:0.05

4 Chapter 4 - Implementation

4.1 Overview

The implementation phase of the ***NeuroAudit*** project focused on developing a fully functional, AI-powered system that can automatically detect security vulnerabilities in source code. Specifically, the system targets two of the most common and dangerous types of attacks: **SQL Injection (SQLi)** and **Cross-Site Scripting (XSS)**. The primary objective was to provide developers and security teams with a smart and accessible tool to identify insecure code before it is deployed into production environments.

The system was implemented using the **Python programming language**, with **TensorFlow** and **Keras** as the core libraries for building and training the deep learning models. The development and experimentation were conducted using **Kaggle Notebooks**, which offered the required computational resources and environment for handling model training tasks.

To provide a practical and user-friendly experience, a **web application** was developed as the main interface for the system. The web app includes:

- A **frontend**, built using **React**, offering a clean, responsive, and interactive user interface where users can input code snippets and view vulnerability detection results in real-time.
- A **backend** developed using a combination of **Node.js** with the **Express** framework and **Flask**. Express handles the server-side logic and routes between the frontend and backend services, while Flask is used to serve the AI models and handle inference requests. Together, they ensure smooth communication between the interface and the deep learning models.
- .

The main components of the system include:

- **Data Preprocessing Module:** Handles data cleaning, tokenization, padding, and formatting of code samples to prepare them for training and evaluation.
- **Model Training and Evaluation:** Three separate deep learning models were implemented:
  - A binary classifier for detecting **SQL Injection**.
  - A binary classifier for detecting **XSS**.
  - A multi-class classifier for identifying whether code is **safe**, **vulnerable to SQLi**, or **vulnerable to XSS**.

- **Prediction Engine:** Exposes the trained models through the backend, enabling real-time classification of submitted code.
- **Web Interface:** Provides an integrated platform for users to interact with the models, submit code for analysis, and receive immediate feedback on its security status.

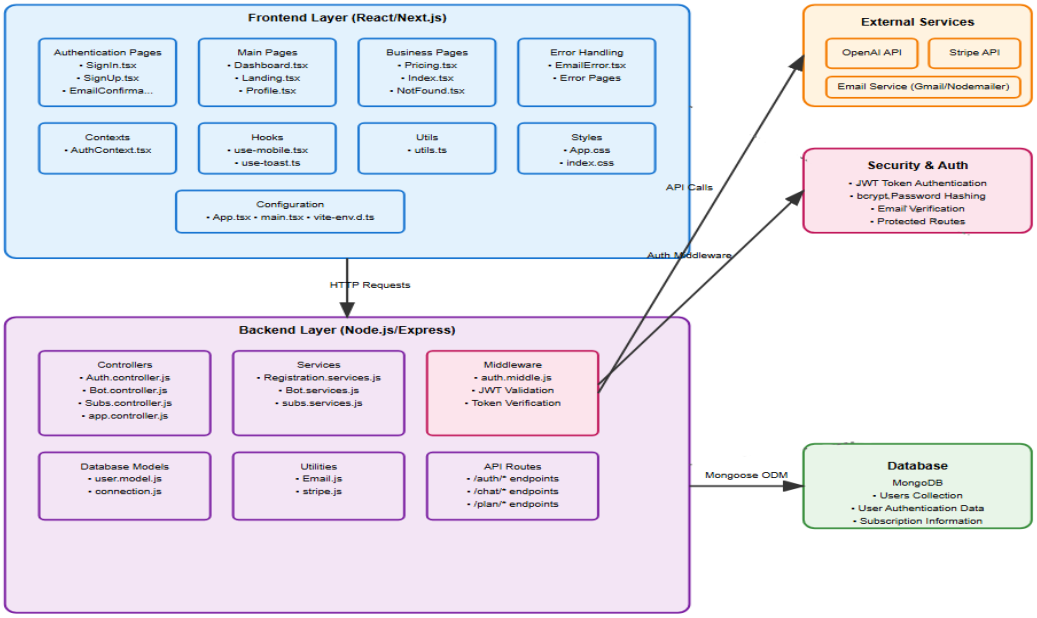
This chapter documents the full implementation journey, detailing the structure, development tools, and key decisions behind each component of the ***NeuroAudit*** system.

4.2 System Architecture Overview

The system follows a modern, microservices-inspired architecture with clear separation of concerns:

- **Frontend Layer:** React-based user interface for code input and results display
- **API Gateway:** Express.js server handling authentication, routing, and business logic
- **AI Processing Layer:** vulnerability detection and OpenAI GPT-4 integration for vulnerability analysis
- **Data Layer:** MongoDB for persistent storage of users, subscriptions, and analysis results
- **External Services:** Stripe for payments, email services for notifications

Block Diagram:



4.3 Data Preprocessing

The preprocessing stage plays a crucial role in preparing the raw dataset for effective training and evaluation of the deep learning models. This section outlines the various steps applied to clean, transform, and structure the data to be suitable for binary and multi-class classification of code vulnerabilities.

Dataset Loading and Cleaning

The dataset used for training was loaded from a CSV which contains abstract syntax tree (AST)-based token representations of code snippets labeled as either safe or containing a vulnerability (SQL Injection or Cross-Site Scripting).

To ensure data quality and reduce redundancy, duplicate entries were removed using the drop_duplicates() function. Null or irrelevant columns such as source_code, vuln_type, and vuln_label were also dropped to retain only the necessary information for the modeling process.

Custom Label Encoding

To prepare the dataset for supervised learning, two different label encoding strategies were implemented depending on the classification task:

Multi-Class Classification

A custom encoding function was applied to generate multi-class labels based on two original columns: vuln_label and vuln_type. The encoding logic assigned:

0 for safe code (vuln_label = good)

1 for SQL Injection vulnerabilities (vuln_label = bad and vuln_type = sql)

2 for Cross-Site Scripting (XSS) vulnerabilities (vuln_label = bad and vuln_type = xss)

This transformed label was stored in a new column named custom_label. To support training of a multi-class classification model, the custom_label column was subsequently **one-hot**

encoded using pd.get_dummies(). This converted the integer class labels into a binary matrix suitable for categorical cross-entropy loss and softmax activation.

Binary Classification

For specialized binary classifiers targeting specific vulnerability types, a simplified label encoding was used. For example, in the sqli binary classifier:

The dataset was filtered to only include rows where vuln_type = sql.

The vuln_label column was directly mapped to integers:

0 for safe code (vuln_label = good)

1 for vulnerable code (vuln_label = bad)

This binary encoding enabled the use of sigmoid activation with binary cross-entropy loss. The filtered and labeled data was then split into training, validation, and test sets with stratification to maintain class balance.

By using separate encoding schemes tailored to the task—multi-class detection vs. targeted binary detection—the models could be optimized more effectively for their specific objectives.

Tokenization and Padding

The tokenized AST representations (stored in the tokens column) were converted into sequences of integers using the Tokenizer class from Keras with the following parameters:

- filters="": disables filtering of characters.
- lower=True: converts all tokens to lowercase.
- split=' ': uses space as the token separator.
- oov_token="<UNK>": reserves a token for unknown words.

The tokenizer was fit on the training data to learn the vocabulary and then used to transform both training and test datasets into sequences of word indices.

To ensure uniform input lengths for the neural network, all sequences were padded using pad_sequences():

- Padding type: post-padding.
- Maximum length: set to the length of the longest training sequence.

This produced fixed-size input vectors for each code snippet, enabling consistent input dimensions for the deep learning architecture.

Dataset Splitting

The dataset was initially divided into training and test sets using a 64%/36% split with stratification to maintain class balance. The test set was then further split into validation and test subsets (each receiving 50% of the original test set). This final setup resulted in three separate datasets:

- Training Set: Used to train the model.
- Validation Set: Used to monitor performance during training and apply early stopping.
- Test Set: Used for final evaluation and performance reporting.

Stratified splitting ensured that all three classes (safe, SQLI, and XSS) were proportionally represented in all datasets, which is essential for balanced model training and robust evaluation.

4.4 Model Implementation

Our system consists of three deep learning models tailored to detect source code vulnerabilities:

- Model 1: SQL Injection Detector (Binary Classifier)
- Model 2: XSS Detector (Binary Classifier)
- Model 3: Multi-Class Classifier (Safe / SQLi / XSS)

All three models share the same core architecture and hyperparameters, derived from a grid search conducted on Model 1 (SQLi Detector) to identify the most effective configuration.

4.4.1 Grid search results

To optimize model performance, we conducted a **grid search** over a range of hyperparameter combinations. This process involved systematically training models using different settings and evaluating their performance using key metrics such as **Accuracy**, **F1-Score**, and **Loss** on the main test set as well as two additional evaluation sets (Dataset 69 and Dataset 140).

- **Hyperparameters and Search Space:**

  The grid search focused on the following hyperparameters:

  - **Batch Size:** 64, 128, 256
  - **Learning Rate (LR):** 0.0001, 0.001, 0.01
  - **Number of Neurons (hidden units):** 32, 64

This resulted in a total of **18 unique combinations (3 × 3 × 2)**. Each configuration was trained and evaluated independently, and the results were recorded.

- **Evaluation Metrics**

  For each run, we recorded:

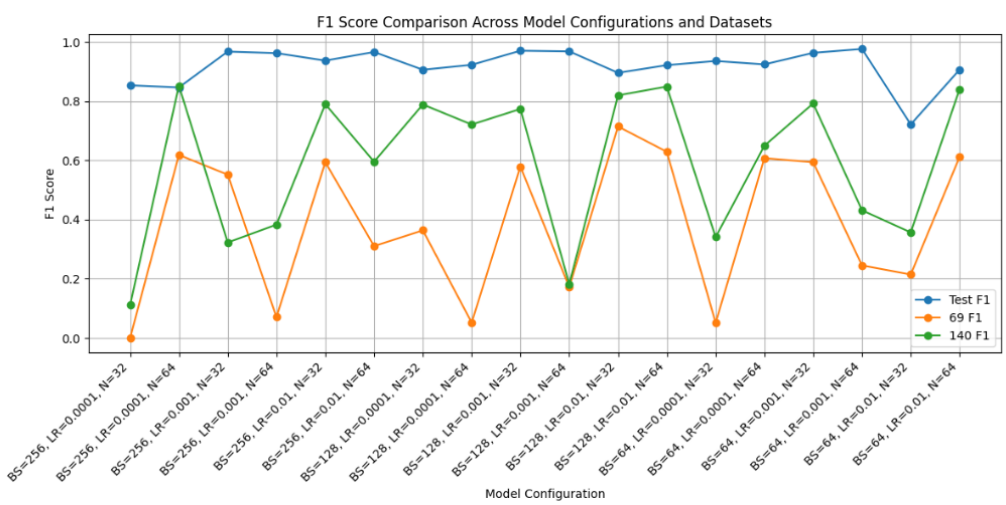
  - **Test Accuracy**, **F1-Score**, and **Loss**
  - **Dataset A** performance: Accuracy, F1, Loss
  - **Dataset B** performance: Accuracy, F1, Loss

  These additional datasets simulate real-world deployment or cross-domain evaluation, allowing us to assess the model's **generalization** capabilities.

- **Results Visualization**

The performance metrics across all configurations are visualized ,which highlights how different parameter combinations affected model performance. This helped us identify optimal trade-offs between performance and generalization.

	Batch Size	LR	Neurons	Test Acc	Test F1	Test Loss	69 Acc	69 F1	69 Loss	140 Acc	140 F1	140 Loss
0	256	0.0001	32	0.942716	0.853351	0.148343	0.437956	0.000000	1.728505	0.314286	0.111111	1.328062
1	256	0.0001	64	0.935900	0.846181	0.156010	0.540146	0.618182	0.939221	0.771429	0.851852	0.557010
2	256	0.0010	32	0.986585	0.967297	0.043154	0.525547	0.551724	1.829781	0.400000	0.322581	2.314865
3	256	0.0010	64	0.984809	0.961968	0.035563	0.423358	0.070588	2.420525	0.378571	0.382979	1.948853
4	256	0.0100	32	0.974331	0.936820	0.064976	0.540146	0.593548	4.652956	0.707143	0.789744	2.506711
5	256	0.0100	64	0.985715	0.965840	0.033452	0.416058	0.310345	4.332488	0.514286	0.595238	2.884881
6	128	0.0001	32	0.962367	0.905893	0.104696	0.540146	0.363636	0.913788	0.735714	0.788571	0.487889
7	128	0.0001	64	0.968385	0.922503	0.074533	0.467153	0.051948	2.600781	0.657143	0.720930	1.304274
8	128	0.0010	32	0.987492	0.970174	0.027057	0.532847	0.578947	2.341009	0.707143	0.773481	1.378687
9	128	0.0010	64	0.986875	0.967987	0.030911	0.437956	0.172043	2.623924	0.292857	0.181818	2.475501
10	128	0.0100	32	0.953049	0.895506	0.113884	0.708029	0.714286	2.664279	0.735714	0.819512	1.712639
11	128	0.0100	64	0.965811	0.921436	0.069802	0.569343	0.628931	3.597943	0.771429	0.849057	1.752998
12	64	0.0001	32	0.974875	0.935469	0.068498	0.459854	0.051282	1.712540	0.364286	0.340741	1.753014
13	64	0.0001	64	0.967950	0.923977	0.083507	0.613139	0.607407	1.129567	0.614286	0.649351	0.816362
14	64	0.0010	32	0.985135	0.962976	0.035252	0.540146	0.593548	1.867483	0.700000	0.792079	1.800706
15	64	0.0010	64	0.990247	0.976138	0.020349	0.416058	0.245283	2.219247	0.378571	0.431373	1.901664
16	64	0.0100	32	0.908056	0.721257	0.158676	0.518248	0.214286	2.288344	0.407143	0.356589	1.373133
17	64	0.0100	64	0.959684	0.906208	0.094621	0.554745	0.611465	3.726911	0.750000	0.840183	0.993856



4.4.2 Model Architecture (Shared across all models)

Each model is a hybrid CNN-LSTM neural network designed to capture both local token patterns and long-range dependencies:

```
model = Sequential([
    Embedding(input_dim=..., output_dim=128),
    Conv1D(filters=64, kernel_size=3, padding='same', activation='relu'),
    Bidirectional(LSTM(units=32, return_sequences=True)),
    GlobalMaxPooling1D(),
    Dropout(0.5),
    Dense(32, activation='relu'),
    Dense(output_units, activation=output_activation)
])
```

- Embedding Layer (128 dim): Converts tokens into dense vectors.
- Conv1D Layer (64 filters): Extracts local features from token sequences.
- Bidirectional LSTM (32 units): Captures sequential context from both directions.
- Global Max Pooling: Reduces sequence output to fixed-size representation.
- Dropout (0.5): Mitigates overfitting.
- Dense (ReLU): Adds non-linearity and learning capacity.
- Output Layer:
  - Sigmoid for binary classifiers (Models 1 & 2).
  - Softmax for multi-class classifier (Model 3).

- Activation Functions and Justification
  - ReLU: Chosen for hidden layers due to its efficiency and ability to mitigate vanishing gradients.
  - Sigmoid: Used for binary output in Models 1 and 2.
  - Softmax: Used in Model 3 for multi-class classification (safe, SQLi, XSS).

- Loss Function, Optimizer, and Metrics
  - Loss Function:
    - binary_crossentropy for Models 1 & 2.
    - categorical_crossentropy for Model 3.
  - Optimizer: Adam optimizer was selected for its adaptive learning rate and efficient convergence behavior.
  - Metrics:
    - Accuracy
    - Custom F1 Score: Implemented as a custom metric to handle class imbalance and reflect the real-world cost of false negatives.
    - Precision
    - Recall
    - loss

- Hyperparameter Tuning (Based on Grid Search on SQLi Detector)

Hyperparameter	Selected Value	Reason
Batch Size	128	Balanced memory usage and model convergence.
Learning Rate	0.01	Provided optimal balance between learning speed and stability.
LSTM Units	32	Enough capacity to model code semantics without overfitting.
Epochs	4	Early stopping prevented overfitting; this range was effective.
Validation Split	0.18	Ensures proper validation during training.

These optimal values were identified via grid search over various combinations and evaluated across three datasets using F1-score.

- Model Summary (Example: SQL Injection Detector)

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 101, 128)	23,808
conv1d (Conv1D)	(None, 101, 64)	24,640
bidirectional (Bidirectional)	(None, 101, 128)	66,048
global_max_pooling1d (GlobalMaxPooling1D)	(None, 128)	0
dropout (Dropout)	(None, 128)	0
dense (Dense)	(None, 64)	8,256
dense_1 (Dense)	(None, 1)	65

Total params: 368,453 (1.41 MB)
Trainable params: 122,817 (479.75 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 245,636 (959.52 KB)

Note: The vocabulary size for Embedding is dataset-dependent.

5 Chapter 5 - Testing and evaluation

5.1 Objectives of Evaluation

The primary objective of this chapter is to systematically assess the performance and reliability of the proposed deep learning models used for vulnerability detection in source code. These models include:

- **Model 1**: SQL Injection Detector (Binary Classifier)
- **Model 2**: XSS Detector (Binary Classifier)
- **Model 3**: Multi-Class Classifier (Safe / SQLi / XSS)

The evaluation process is essential for determining how well these models fulfill their intended purpose—accurately identifying security vulnerabilities in code snippets—and ensuring they generalize effectively to unseen or real-world data.

- **What We Are Testing**

The evaluation focuses on several key aspects:

- **Accuracy**: How often the models correctly predict the vulnerability class.
- **F1-Score**: A balanced metric combining precision and recall, particularly important in imbalanced datasets where false negatives (e.g., undetected vulnerabilities) are critical.
- **Generalization**: The ability of each model to maintain strong performance on external test datasets not seen during training.
- **Robustness**: How stable and reliable the predictions are across varying data distributions and real-world code samples.

Types of Evaluation Used

To thoroughly evaluate the models, we employ a combination of **quantitative** and **qualitative** evaluation methods:

- **Quantitative Metrics**:
  Standard classification metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix are used to measure model performance. These metrics are computed on both the internal validation split and multiple external test datasets.

- **Cross-Dataset Evaluation**:
  Models are tested not only on held-out validation sets but also on independent datasets containing samples from different sources, to assess how well they generalize beyond the training data.
- **Visual Analysis**:
  We use plots such as bar charts (for class-wise performance), ROC curves (to visualize trade-offs between true positive and false positive rates), and heatmaps of normalized confusion matrices to qualitatively understand the models' behavior.

Through this evaluation, we aim to determine the effectiveness and limitations of each model and identify potential areas for future improvement or enhancement.

5.2 Evaluation Setup

Dataset Splitting

To ensure effective training and unbiased evaluation, the primary dataset was divided into three parts:

- 46% for training the models.
- 18% for validation during training.
- 36% for testing final performance.

The split was performed using stratified sampling, which preserves the original distribution of vulnerable vs. safe code across all subsets. This helps maintain balance and consistency, especially important in security-related classification tasks where vulnerable examples may be less frequent.

The training set was used to fit the model weights, the validation set was used to tune hyperparameters and apply early stopping (to prevent overfitting), and the test set remained completely unseen until the final evaluation phase.

This splitting strategy ensures a robust performance estimate, while allocating a sufficient amount of data for validation and final testing.

Out-of-sample Testing

To assess the models' ability to **generalize beyond the training data** and simulate real-world deployment conditions, we evaluated them using two specially constructed external datasets:

- **Dataset A**
  - **Composition**: 140 manually curated PHP code samples, evenly split between **secure** and **SQL Injection (SQLi)-vulnerable** snippets.
  - **Purpose**: This dataset was designed to provide a realistic, out-of-distribution evaluation of the **binary SQLi classifier**. It mimics the unpredictability and diversity of real-world code encountered outside of controlled datasets.
- **Dataset B**
  - **Composition**: 69 real-world examples of SQLi-vulnerable code and 69 **hand-sanitized**, secure counterparts.
  - **Purpose**: Used to test the model's **sensitivity to input sanitization techniques**. This dataset helps verify that the model can detect nuanced changes that distinguish insecure code from its secured versions, validating its **discriminative power** and **robustness**.

These two datasets act as **stress tests** for both the binary and multi-class models. Their design helps ensure that the classifiers are not merely memorizing patterns in the training set but are learning to generalize to **unseen, structurally diverse** inputs — a crucial requirement for effective deployment in real-world code auditing scenarios.

5.3 Model Performance

This section presents a comprehensive evaluation of the three deep learning models:

- Model 1: SQL Injection Detector (Binary Classifier)
- Model 2: XSS Detector (Binary Classifier)
- Model 3: Multi-Class Vulnerability Classifier (Safe / SQLi / XSS)

Each model was evaluated using four different setups:

1. Training Set (46% of original data)
2. Validation Set (18% of original data, used during training)
3. Test Set (36% of original data)
4. Out-of-Sample Datasets(for models 1&3):
    - Dataset A: 140 real PHP snippets (balanced SQLi and safe)
    - Dataset B: 69 vulnerable vs 69 secure versions of the same samples

We report key performance metrics: accuracy, F1-score, precision, recall, and loss, along with visualizations for model diagnostics and generalization analysis.

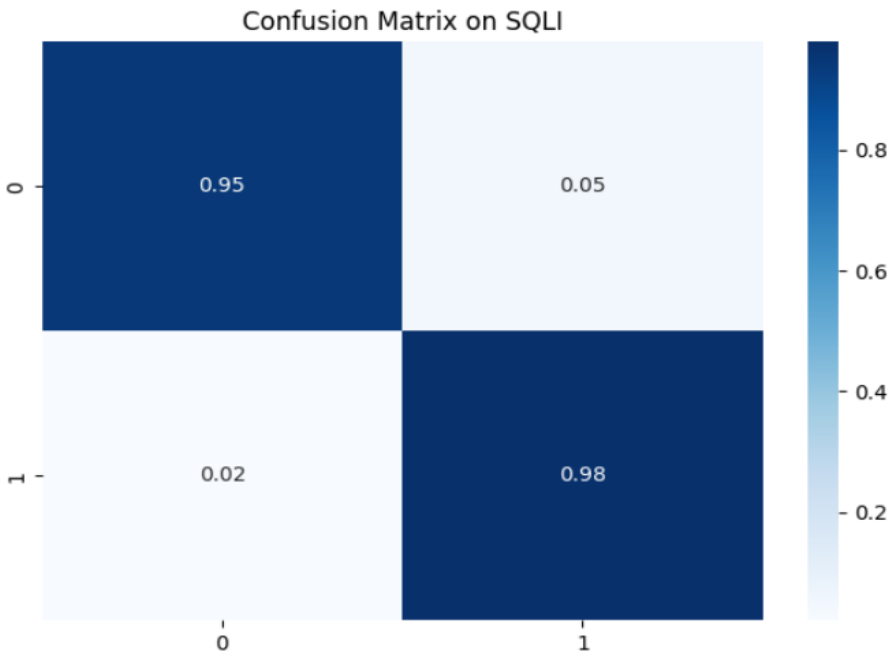
Model 1: SQLi

Results:

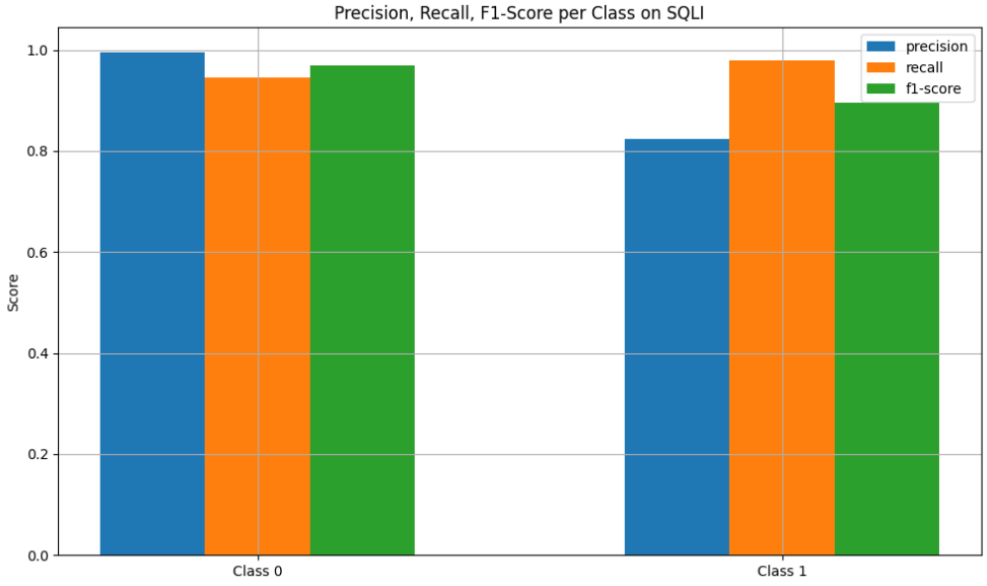
Evaluation Set	Accuracy	Precision	Recall	F1-Score	Loss
Train	95%	83%	98%	90%	0.11
Test	95%	82%	98%	90%	0.11
Dataset A	74%	79%	85%	82%	1.7
Dataset B	72%	70%	72%	71%	2.66

Observation: High training and test performance indicate effective learning, while noticeable drops on Datasets A and B highlight reduced generalization on real-world and sanitized code samples.

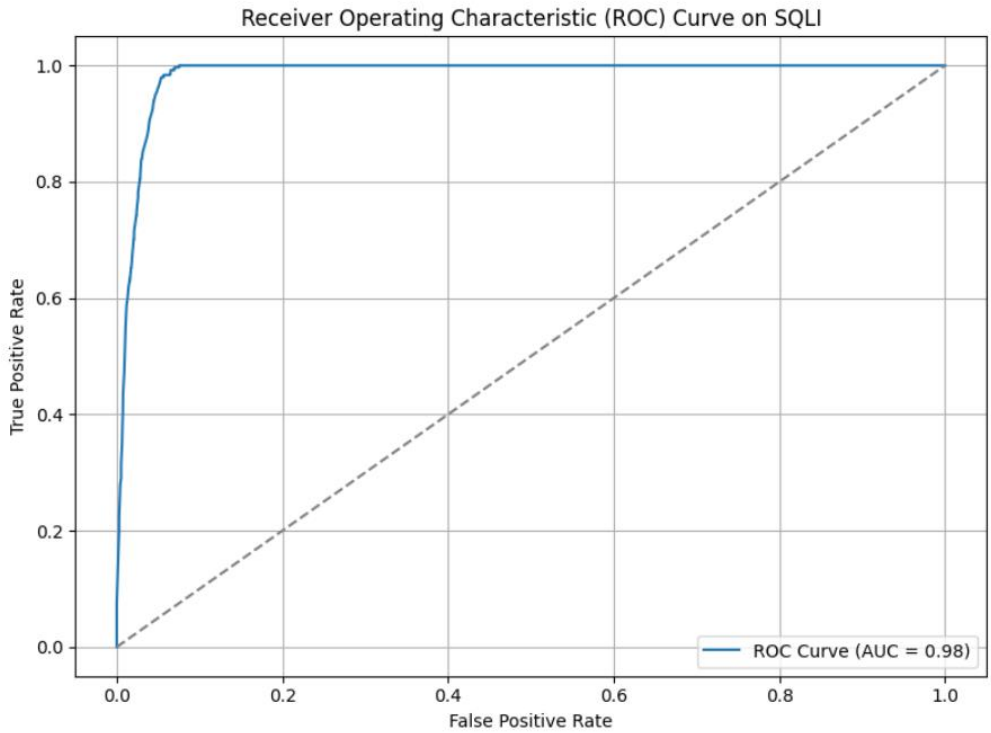
Confusion Matrix on Test Set:



Bar chart Precision, Recall, F1-Score per Class:



Receiver Operating Characteristic (ROC) Curve:



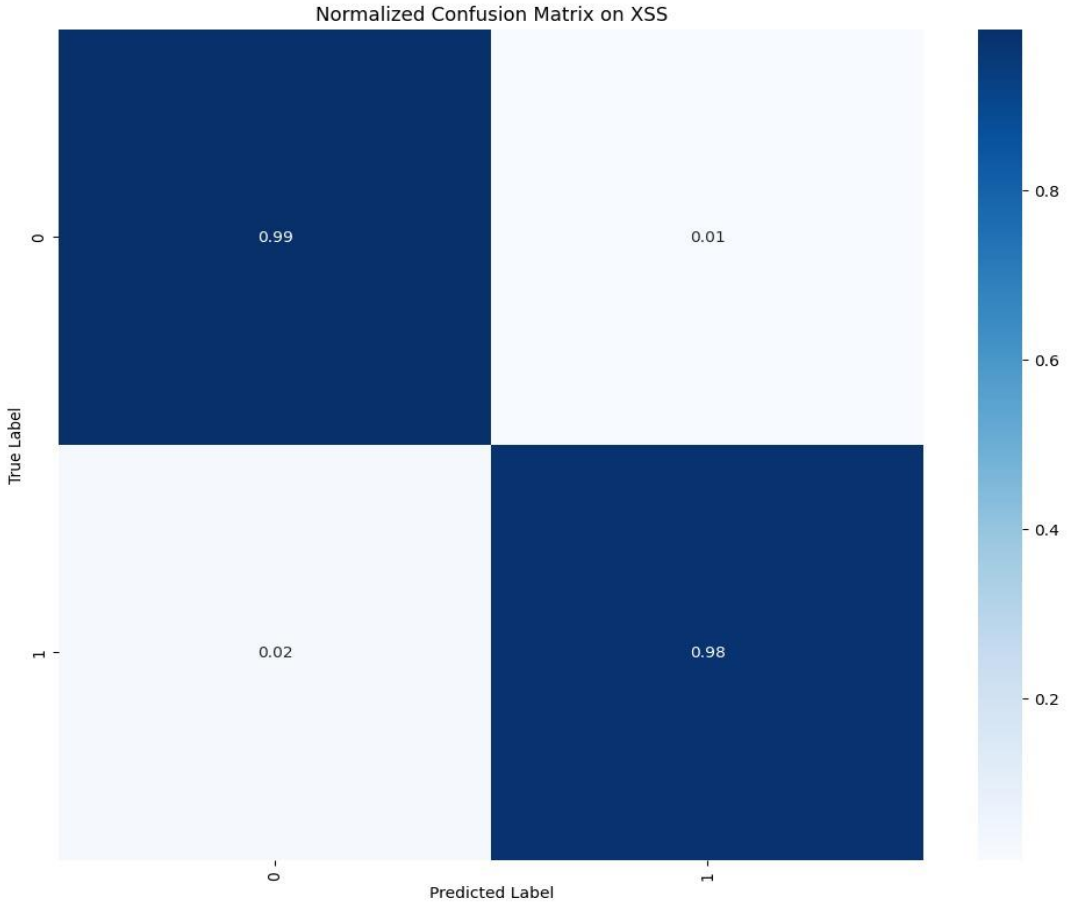
Model 2: XSS

Results:

Evaluation Set	Accuracy	Precision	Recall	F1-Score	Loss
Train	99%	95%	98%	97%	0.018
Validation	99%	95%	98%	96%	0.02
Test	99%	95%	98%	96%	0.02

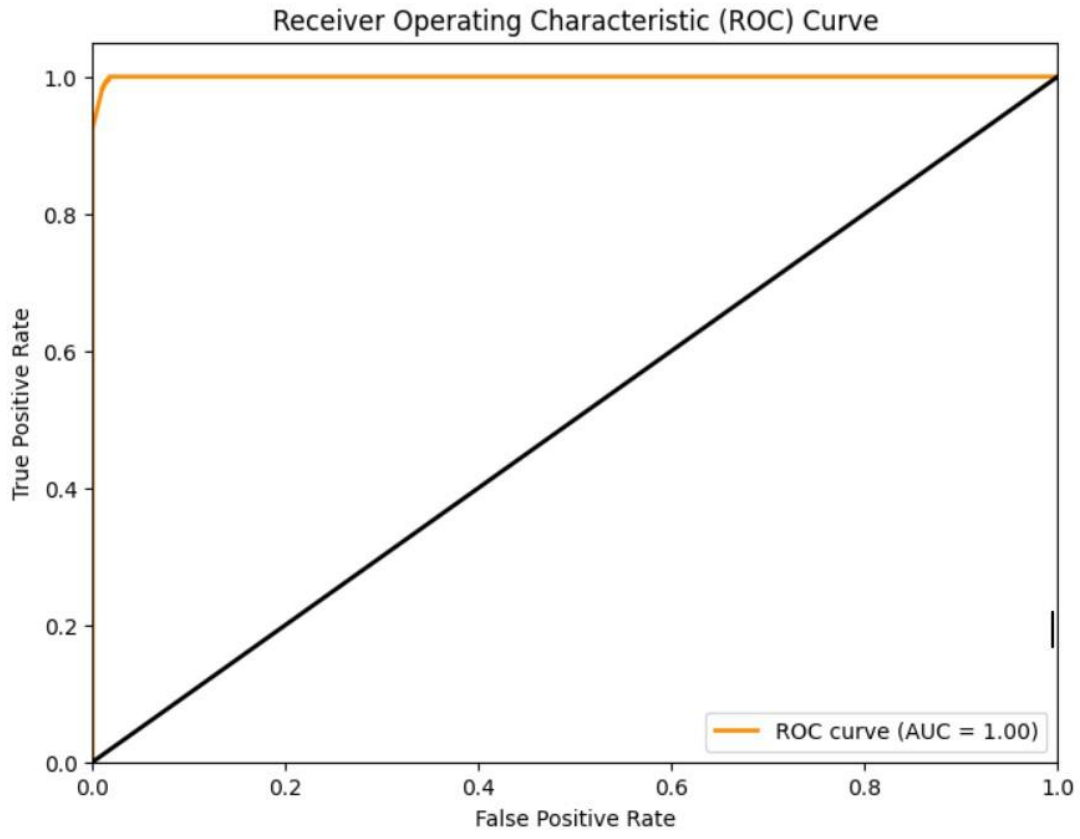
Observation: The XSS model achieved consistently strong performance across all evaluation sets, with no significant signs of overfitting or underfitting.

Confusion Matrix on Test Set:



Bar chart Precision, Recall, F1-Score per Class:

Receiver Operating Characteristic (ROC) Curve:



Model 3: Multi-Class

Results:

Evaluation Set	Accuracy	Precision(macro)	Recall(macro)	F1-Score	Loss
Train	97%	92%	98%	94%	0.05
Validation	97%	92%	98%	94%	0.05
Test	97%	92%	98%	95%	0.05
Dataset A	76%	49%	48%	48%	------
Dataset B	54%	39%	36%	37%	------

Observation: The model shows excellent in-sample performance. However, it exhibits sharp performance declines on Dataset A and Dataset B. This drop is primarily due to the fact that both datasets contain only SQL Injection (SQLi) samples, which limits the diversity of attack types. The model, trained on a balanced mix of classes (Safe, SQLi, XSS), may struggle to generalize when evaluated on datasets with a single attack type, indicating limited robustness to skewed or out-of-distribution inputs.

Evaluation Set	Precision(weighted)	Recall(weighted)	F1-Score
Dataset A	78%	76%	77%
Dataset B	58%	54%	56%

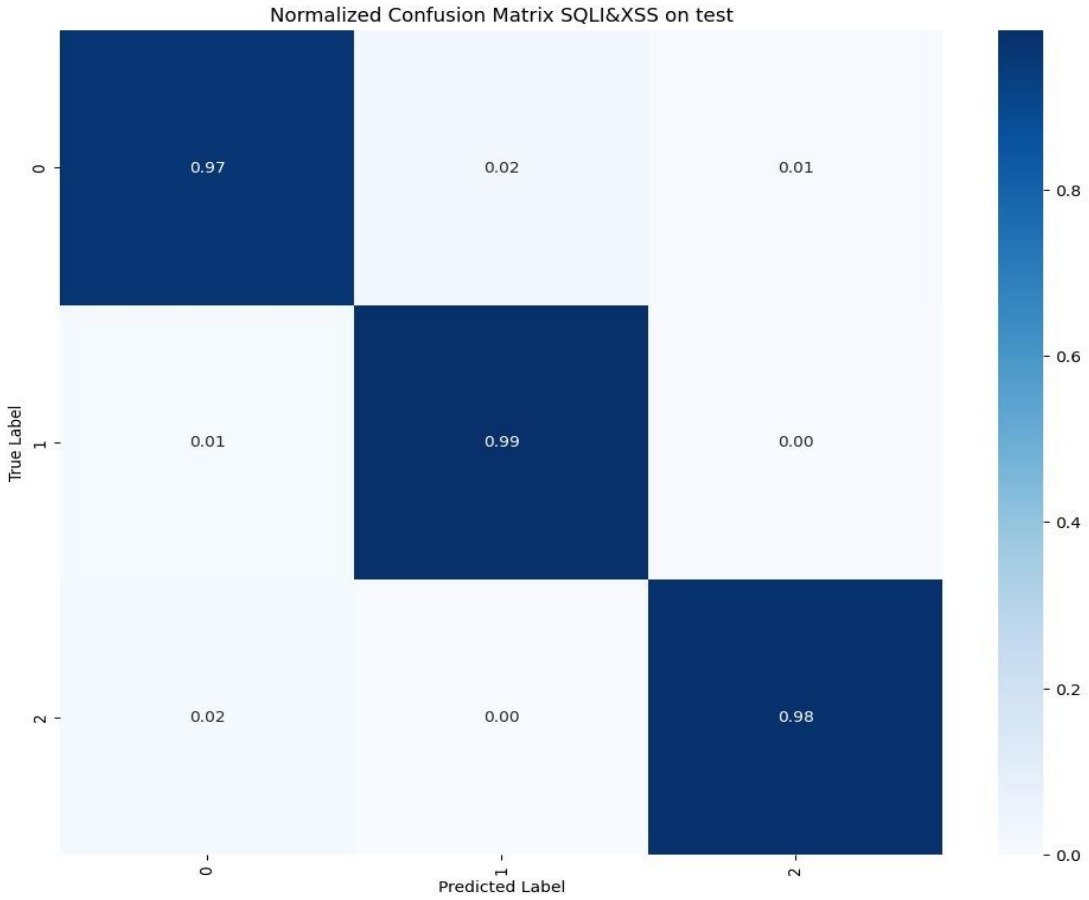
Observation: While the multi-class model demonstrates **strong weighted performance on the internal test set**, its effectiveness significantly drops when evaluated on **Dataset A (F1-score: 77%)** and even more on **Dataset B (F1-score: 56%)**. This decline reflects a **reduced generalization capacity** when exposed to real-world or imbalanced conditions.

The primary reason for this drop is that both Dataset A and Dataset B contain **only SQL Injection (SQLi) and secure samples**, **excluding XSS examples**. This leads to a **skewed class distribution**, causing the model to misclassify due to **underexposure to similar out-of-distribution scenarios** during

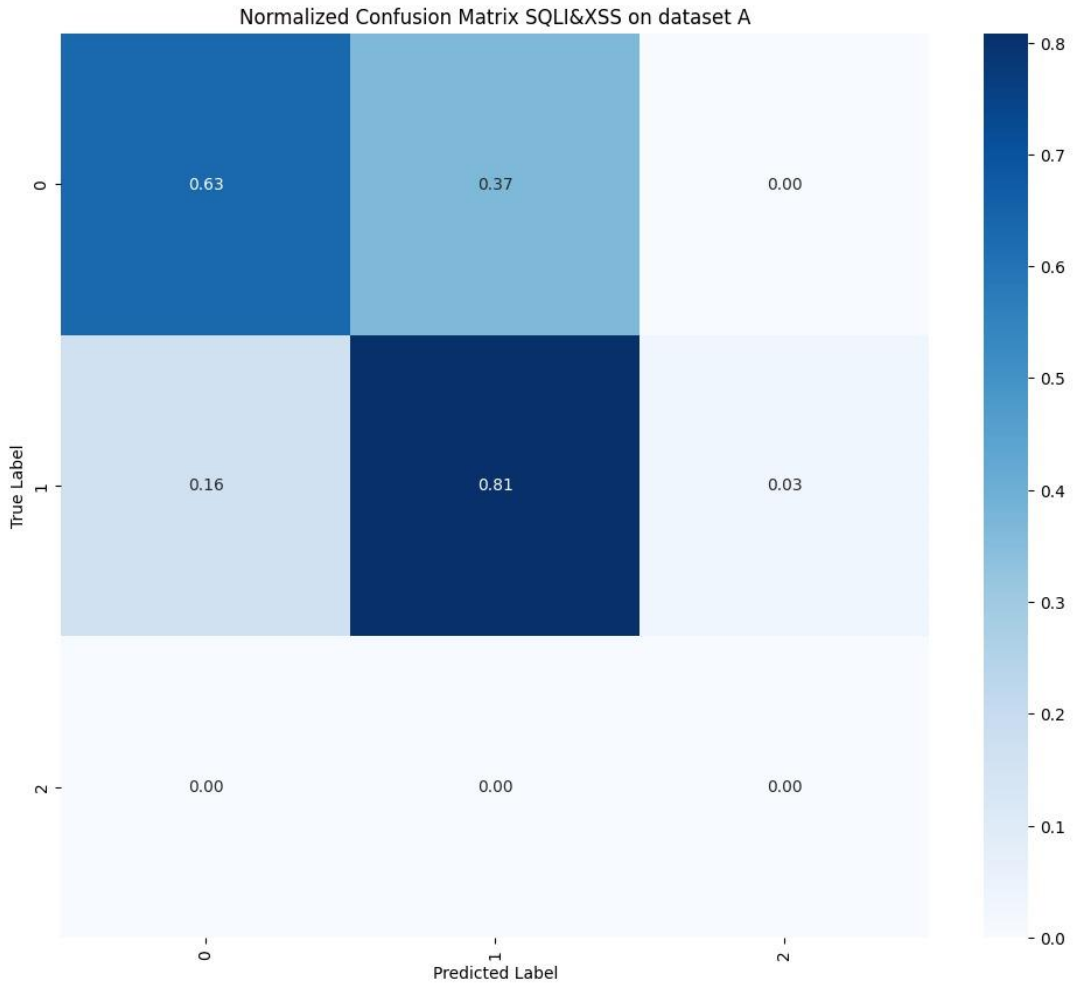
training. Consequently, the weighted metrics—while typically useful for assessing overall performance—become **less indicative of true robustness** in the presence of **class imbalance** and **distribution shifts**.

These results emphasize the importance of evaluating machine learning models not just on clean, balanced datasets but also on **unbalanced and adversarial samples that simulate real-world input**, to fully understand their limitations and areas for improvement.

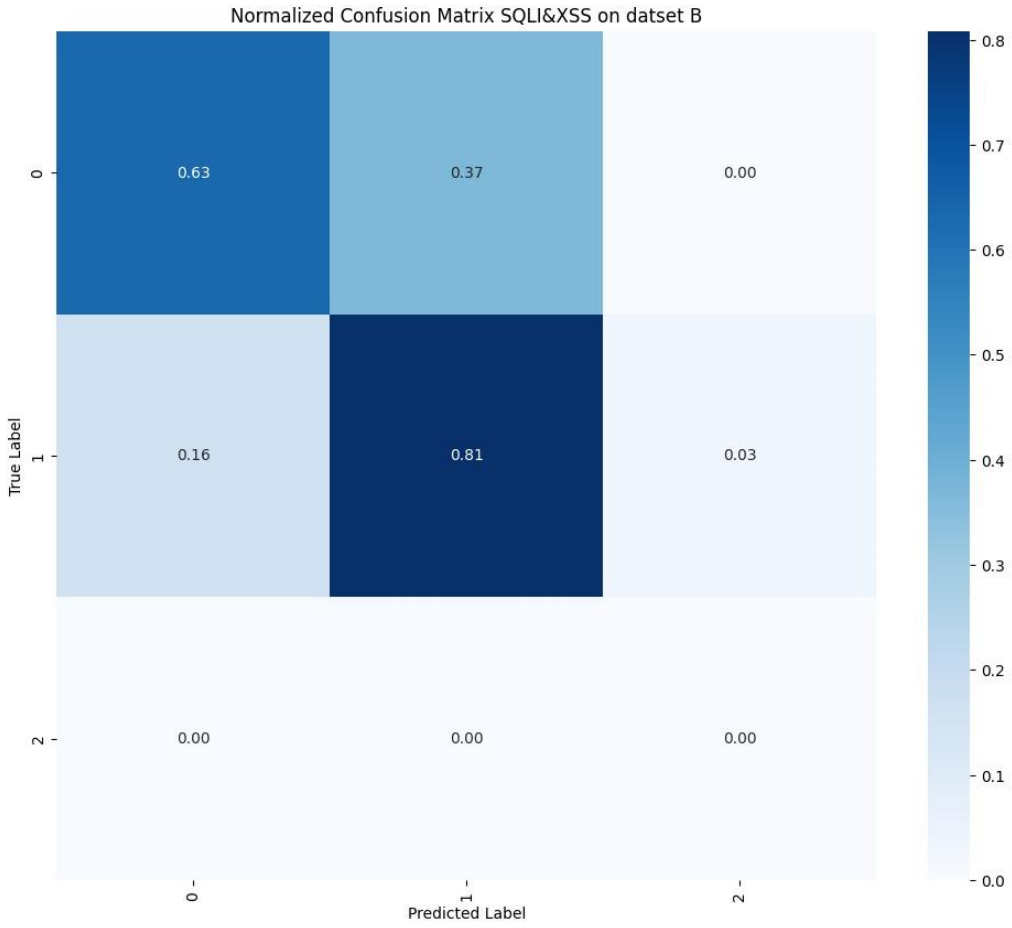
Confusion Matrix on Test Set:



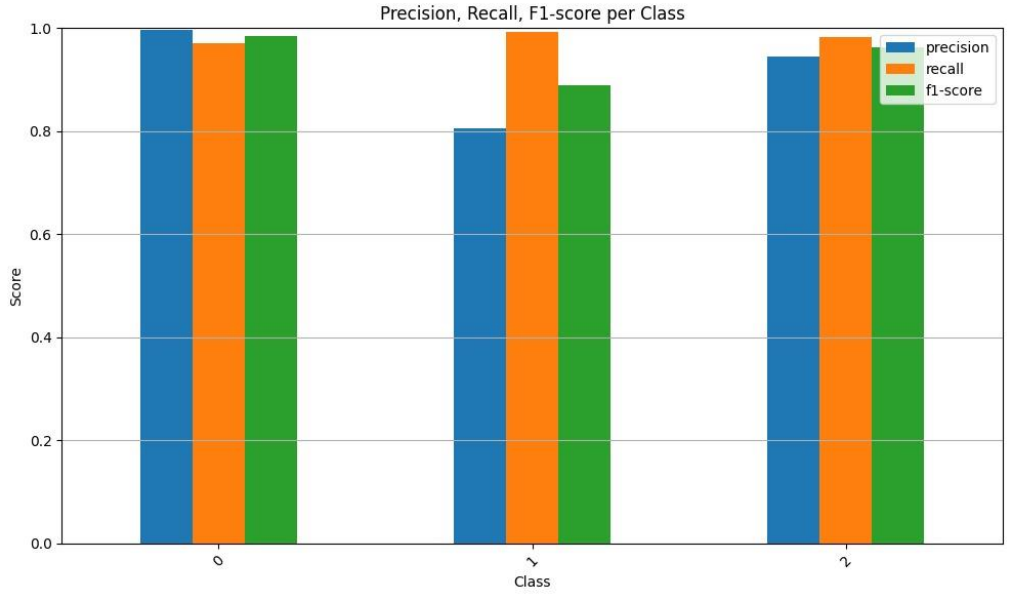
Confusion Matrix on dataset A:



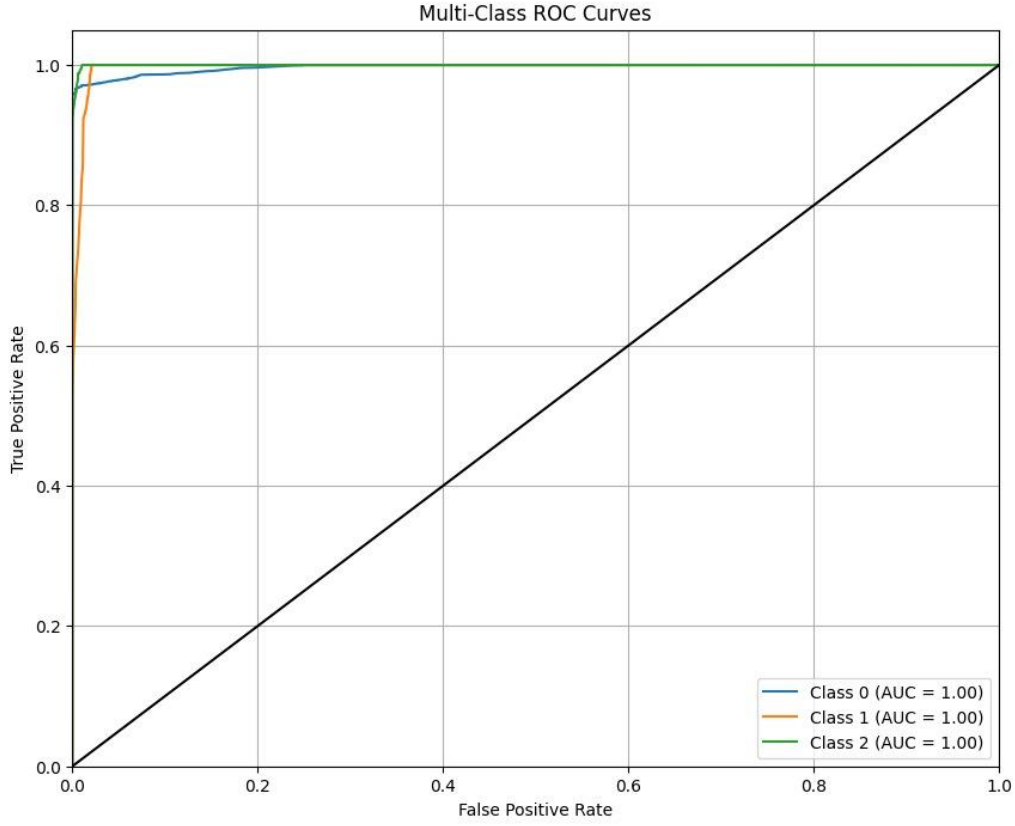
Confusion Matrix on dataset B:



Bar chart Precision, Recall, F1-Score per Class:



Receiver Operating Characteristic (ROC) Curve:



6 Chapter 6 - Results and Discussions

6.1 Introduction

This chapter presents a detailed analysis of the performance outcomes for the three models developed in this study: the **SQL Injection (SQLi) binary classifier**, the **Cross-Site Scripting (XSS) binary classifier**, and the **multi-class classifier** that distinguishes between safe code, SQLi, and XSS vulnerabilities.

Building upon the evaluation framework outlined in Chapter 5, this chapter interprets the models' behavior across different evaluation scenarios, including standard training, validation, and test splits, as well as two **out-of-sample datasets** specifically designed to assess generalization, robustness, and real-world applicability. These additional datasets (Dataset A and Dataset B) contain manually curated and sanitized PHP code samples that simulate practical web application threats.

The primary goals of this chapter are to:

- **Analyze model performance metrics** such as accuracy, precision, recall, F1-score, and loss.
- **Compare model behavior** across in-sample and out-of-sample datasets.
- **Interpret generalization ability**, robustness against code sanitization, and the potential for overfitting or underfitting.
- **Discuss trade-offs and challenges** observed during testing, especially in handling realistic and previously unseen code patterns.

The discussions in this chapter aim to draw meaningful insights from the performance data, identify strengths and limitations of each model, and suggest directions for future improvements.

6.2 Summary of Results

This section summarizes the key evaluation outcomes for the three developed models **SQLi binary classifier**, **XSS binary classifier**, and the **multi-class vulnerability classifier** as previously presented in Chapter 5. The performance metrics include **accuracy**, **precision**, **recall**, **F1-score**, and **loss**, reported across multiple evaluation settings: training, validation, testing, and two out-of-sample datasets (Dataset A and Dataset B).

Model	Domain	Accuracy (Test)	Accuracy (Dataset A)	Accuracy (Dataset B)
SQLi	Binary	95%	74%	72%
XSS	Binary	99%	----------	----------
Multi-Class	3-Class	97%	76%	54%

6.3 Analysis of Model Performance

- The XSS model demonstrates outstanding and consistent performance, achieving 99% accuracy across training, validation, and test sets. Its high precision, recall, and F1-score indicate excellent generalization and reliable detection of XSS vulnerabilities, even without separate out-of-sample testing.
- The SQLi model, evaluated separately to gain deeper insight, shows strong accuracy (95%) on both training and test sets with high F1-score and low loss. When tested on realistic, unseen data (Dataset A and B), it maintains solid performance (74% and 72% accuracy), confirming its ability to generalize to real-world code, including sanitized inputs that challenge traditional detection methods.
- The multi-class model, which is the primary focus of this project, achieves 97% accuracy on both training and test sets, successfully distinguishing between safe code, SQLi, and XSS vulnerabilities. Its balanced macro-average metrics reflect strong learning across all classes. On out-of-sample datasets, despite natural class imbalance, it reaches 76% and 54% accuracy, showing it can still detect SQLi-related patterns under domain shift.
- In summary, the models, especially the multi-class classifier, exhibit high effectiveness in vulnerability detection. The binary models provided valuable insights into per-vulnerability performance, and the multi-class model offers a strong, scalable solution for secure

code classification. These results support the model's practical viability while also guiding future improvements in handling diverse real-world code scenarios.

6.4 Limitations

The constraints encountered by this research that warrant consideration for future endeavors include:

- NeuroAudit is a PHP code reviewer that examined the **PHP Test Suite – XSS, SQLi 1.0.0 (2022)** dataset, which, however, does not reflect real-world scenarios. Most files were limited to fewer than 50 lines and contained only fragments of vulnerabilities, failing to encompass all potential cases. Furthermore, the conversion of the dataset to Abstract Syntax Trees (ASTs) revealed numerous duplicated files, likely resulting from prior data augmentation. As a result, we sought to identify or construct alternative examples. However, due to time constraints, we collected approximately 200 files designed to represent a range of real-world scenarios.
- In addition, limited computational resources restricted our ability to explore advanced methods such as bytecode analysis, as our hardware was unable to process more than 4,000 samples. Nonetheless, bytecode remains a promising direction for future research due to its potential to provide deeper semantic insights.

6.5 Implications

The results of this work offer several promising implications for the field of automated secure code analysis:

- **Practical Viability**: The promising accuracy and F1-scores of the multi-class classifier on real-world scenarios indicate that hybrid learning model (CNN+LSTM) can effectively support secure code review, particularly for identifying common vulnerabilities like SQL Injection (SQLi) and Cross-Site Scripting (XSS).

- **Complementing Traditional Tools**: This system showcases the potential of machine learning to complement or even extend beyond rule-based and static analysis tools, especially in identifying novel or obfuscated attack patterns.
- **Advancing AI for Code Security**: By integrating deep learning techniques into the vulnerability detection pipeline, this project contributes to the growing body of research exploring how neural models can capture security-relevant code patterns in a scalable and automated fashion.
- **Domain Adaptability**: The binary evaluation of SQLi and XSS provides additional insight into how specialized classifiers can be applied or fine-tuned for domain-specific detection tasks, offering flexibility in practical deployment scenarios.

6.6 Suggestions for Improvement

To build on the current work and address the identified challenges, the following improvements are recommended for future iterations:

- **Larger and More Diverse Datasets**: Expanding the dataset to include longer code snippets, more varied examples, and diverse vulnerability types would improve model generalization and robustness.
- **Adversarial Robustness and Obfuscation Handling**: Future research could focus on making the models more resilient to adversarially crafted or obfuscated inputs, further aligning the system with real-world threat models.
- If there were no limited computational resources restricted try to explore advanced methods such as bytecode analysis, as our hardware was unable to process more than 4,000 samples. Nonetheless, bytecode remains a promising direction for future research due to its potential to provide deeper semantic insights.

7 Chapter 7 – System Analysis & Design

7.1 System Overview

What is NeuroAudit?

NeuroAudit is an intelligent, AI-powered web application designed to revolutionize how developers approach code security. Built on cutting-edge artificial intelligence technology, **NeuroAudit** specializes in detecting and explaining critical security vulnerabilities in PHP applications, with a primary focus on SQL injection and Cross-Site Scripting (XSS) attacks.

Core Functionality

The system leverages a sophisticated AI model, trained specifically on vulnerability patterns and secure coding practices, to provide:

- **Real-time Code Analysis**: Instant vulnerability detection as developers write or paste code
- **Intelligent Pattern Recognition**: Advanced AI algorithms that understand context and complex attack vectors
- **Educational Insights**: Detailed explanations of vulnerabilities with remediation guidance
- **User-Friendly Interface**: Intuitive web-based platform accessible to developers of all skill levels
- **Subscription-Based Access**: Flexible pricing tiers to accommodate different organizational needs

7.2 Motivation

The Growing Cybersecurity Challenge

In today's rapidly evolving digital landscape, web applications face an unprecedented number of security threats. SQL injection and Cross-Site Scripting (XSS) attacks remain among the most prevalent and dangerous vulnerabilities, consistently ranking in the OWASP Top 10 security risks.

7.3 Problem Statement

SQL Injection Vulnerabilities

- **Dynamic Query Construction**: Improper concatenation of user input with SQL queries

- **Inadequate Input Sanitization**: Failure to properly validate and escape user inputs
- **Stored Procedure Misuse**: Incorrect implementation of parameterized queries
- **Second-Order SQL Injection**: Complex vulnerabilities that manifest through data storage and retrieval

Cross-Site Scripting (XSS) Vulnerabilities

- **Reflected XSS:** User input directly reflected in web pages without proper **encoding**
- **Stored XSS:** Malicious scripts permanently stored and executed for multiple users
- **DOM-based XSS:** Client-side script vulnerabilities in dynamic content manipulation
- **Context-Specific Encoding:** Failure to apply appropriate encoding based on output context

Developer Challenges

- **Knowledge Gaps**: Insufficient understanding of security principles and attack vectors
- **Time Constraints**: Pressure to deliver features quickly, often at the expense of security
- **Tool Complexity**: Existing security tools require specialized expertise to use effectively

7.4 Objectives

Intelligent Vulnerability Detection

- Develop an AI-powered system capable of accurately identifying SQL injection and XSS vulnerabilities in PHP code
- Achieve high detection accuracy with minimal false positives through advanced machine learning techniques
- Support multiple vulnerability types including SQL injection, XSS, and complex injection scenarios

User-Friendly Interface

- Create an intuitive web-based platform that requires no specialized security expertise to operate
- Implement real-time code analysis with immediate feedback and suggestions
- Provide clear, actionable reports with vulnerability explanations and remediation guidance
- Enable seamless integration into existing development workflows and processes

7.5 Tools & Technologies

AI/Machine Learning Stack

Custom Model Training

- **Training Dataset: collection of PHP code samples with labeled vulnerabilities**
    - SQL injection examples across different contexts and complexity levels
    - XSS vulnerabilities in various output scenarios
    - Secure code patterns and best practices
    - Real-world vulnerable code from security databases

OpenAI GPT-4: large language model for code analysis and after vulnerability detection

- Advanced natural language processing capabilities
- Code understanding
- Explanation generation for detected issues and provide advices

Backend Technology Stack

Server Infrastructure

- **Node.js: Runtime environment for scalable server-side applications**
    - Event-driven, non-blocking I/O for high performance
    - Rich ecosystem of security and development libraries
- **Express.js: Web application framework**
    - RESTful API development
    - Middleware support for authentication and security
    - Flexible routing and request handling

Database Management

- **MongoDB: NoSQL database for flexible data storage**
  - User account and subscription management
  - Scalable document-based architecture
- **Mongoose ODM: Object Document Mapping for MongoDB**
  - Schema definition and validation
  - Query optimization and data relationships
  - Built-in security features and data sanitization

Authentication & Security

- **JSON Web Tokens (JWT): Secure user authentication**
  - Stateless authentication mechanism
  - Secure token-based session management
  - Integration with subscription management
- **bcrypt: Password hashing and security**
  - Salted password hashing
  - Protection against rainbow table attacks
  - Configurable complexity for future-proofing

Frontend Technology Stack

User Interface Framework

- **React.js: Modern frontend framework**
  - Component-based architecture for maintainable code
  - Virtual DOM for optimal performance
  - Rich ecosystem of UI components and libraries
- **TypeScript: Type-safe JavaScript development**
  - Enhanced code reliability and maintainability
  - Better IDE support and developer experience
  - Compile-time error detection

Development Tools

- **Vite: Modern build tool and development server**
    - Fast hot module replacement for development
    - Optimized production builds
    - Plugin ecosystem for extended functionality

Integration & External Services

Payment Processing

- **Stripe: Subscription and payment management**
    - Secure payment processing with PCI compliance
    - Flexible subscription models and billing cycles
    - Comprehensive API for payment integration

Communication Services

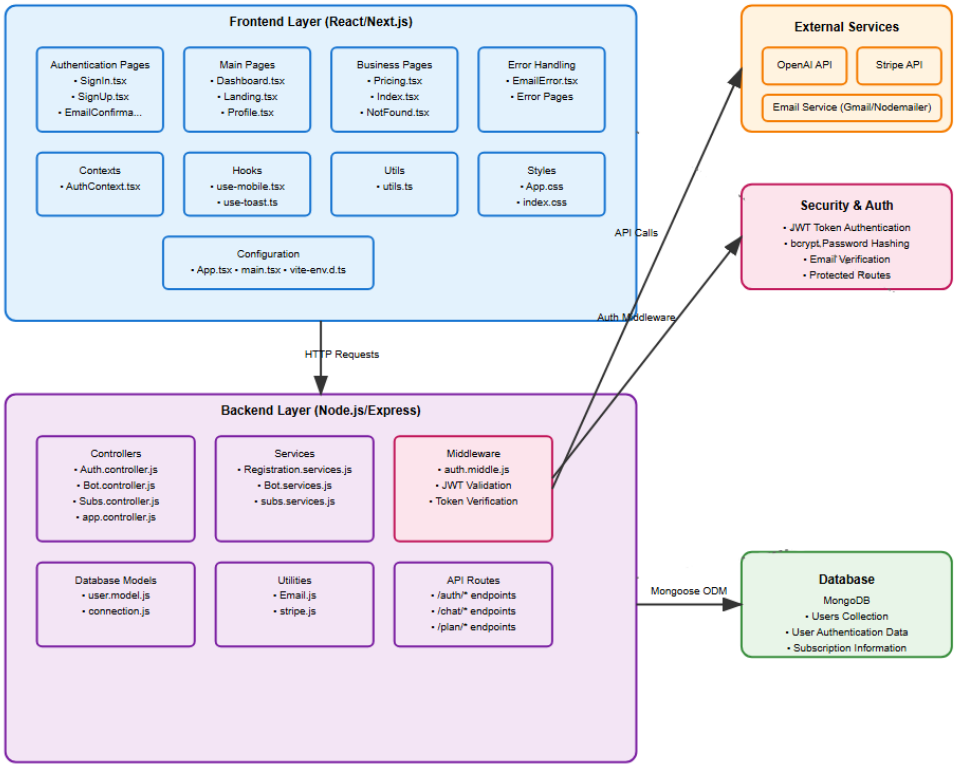
- **Nodemailer: Email service integration**
    - User account verification and notifications
    - Security alerts and vulnerability reports
    - Marketing communications and updates
    - SMTP integration with major email providers

API Integration

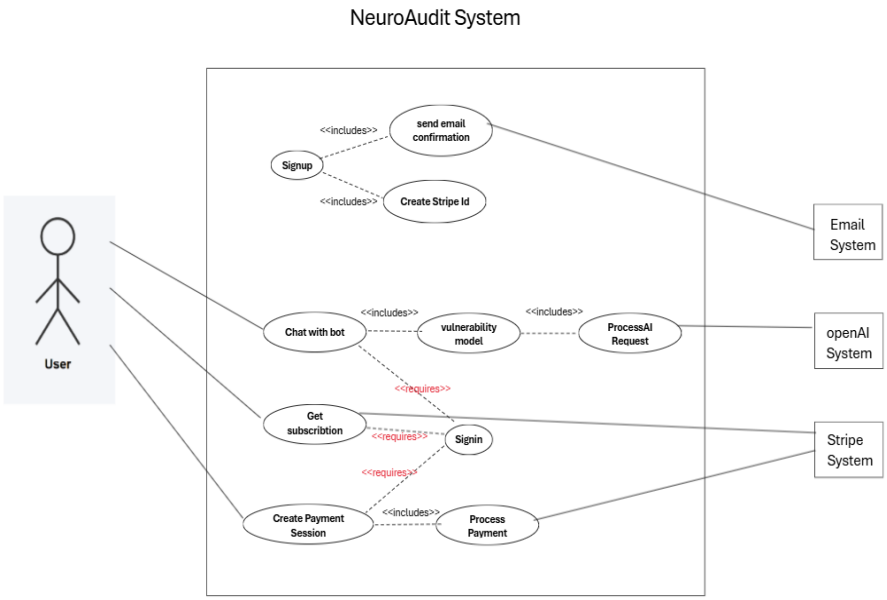
- **OpenAI API: Advanced AI capabilities**
    - Code analysis and vulnerability detection
    - Natural language explanations and guidance
    - Continuous model improvements and updates
    - Scalable API access with usage monitoring

7.6 Diagrams

7.6.1 Block Diagram

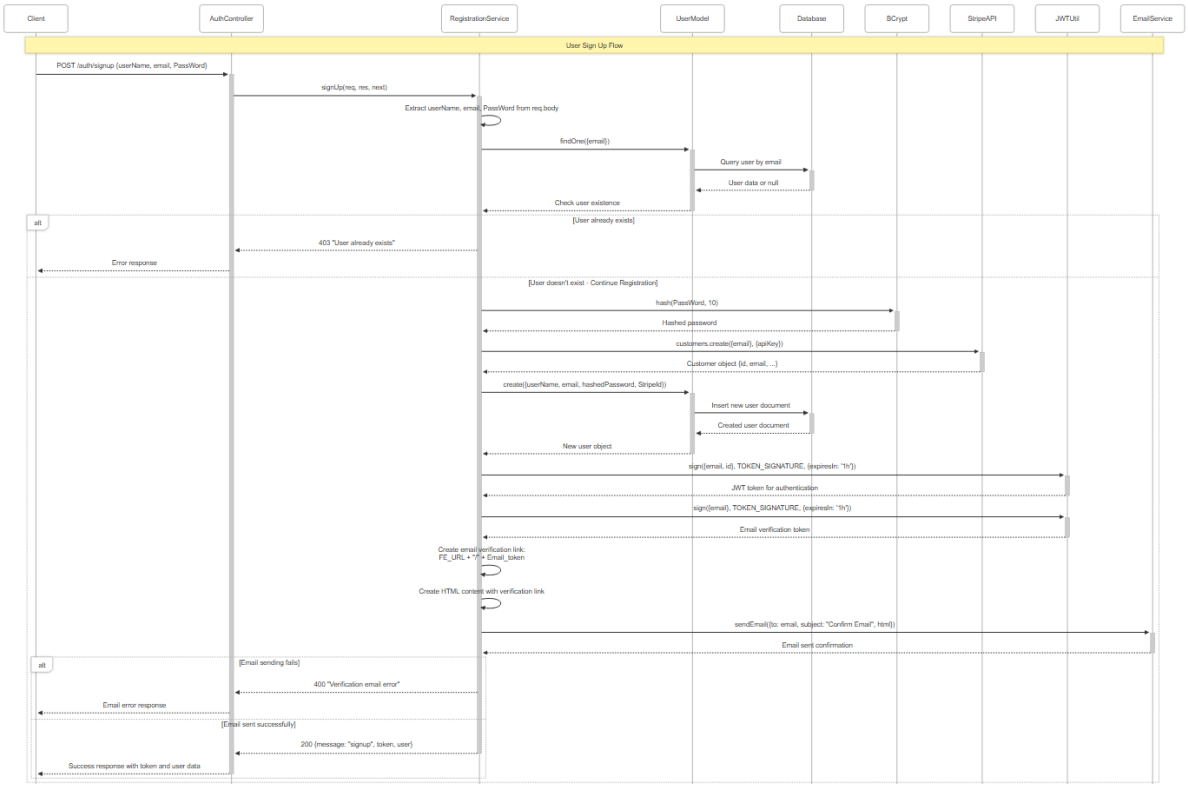


7.6.2 Use case

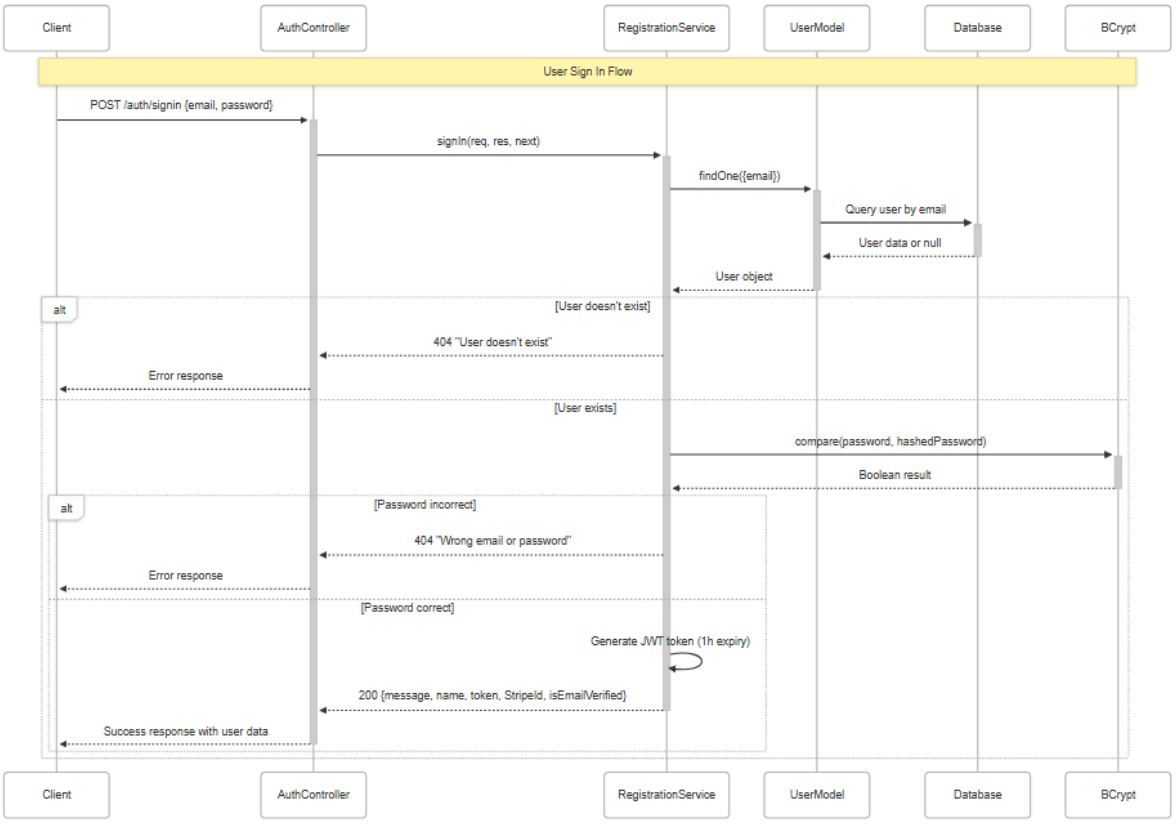

NeuroAudit System
User
Signup
<<includes>>
send email confirmation
<<includes>>
Create Stripe Id
Chat with bot
<<includes>>
vulnerability model
<<includes>>
ProcessAI Request
<<requires>>
Get subscribtion
<<requires>>
Signin
<<requires>>
Create Payment Session
<<includes>>
Process Payment
Email System
openAI System
Stripe System

7.6.3 Sequence diagrams

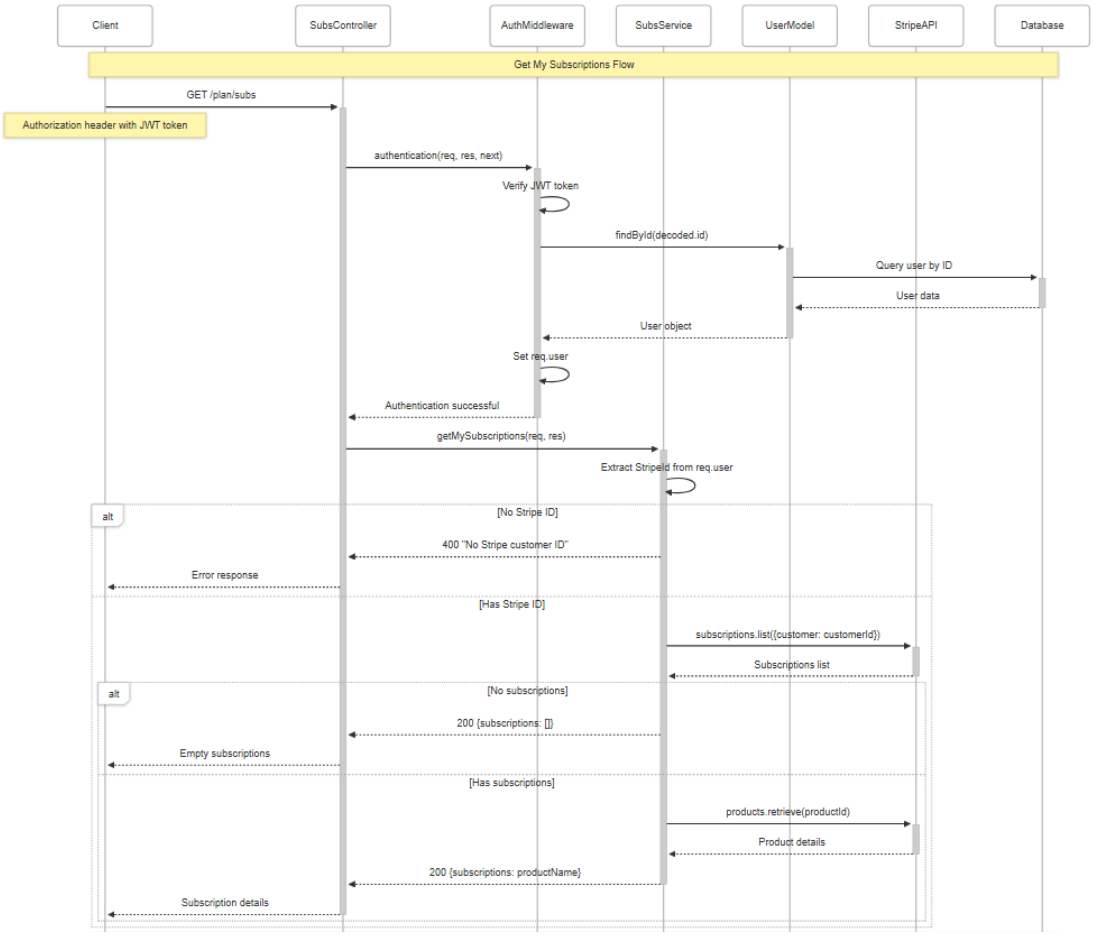
Sign up



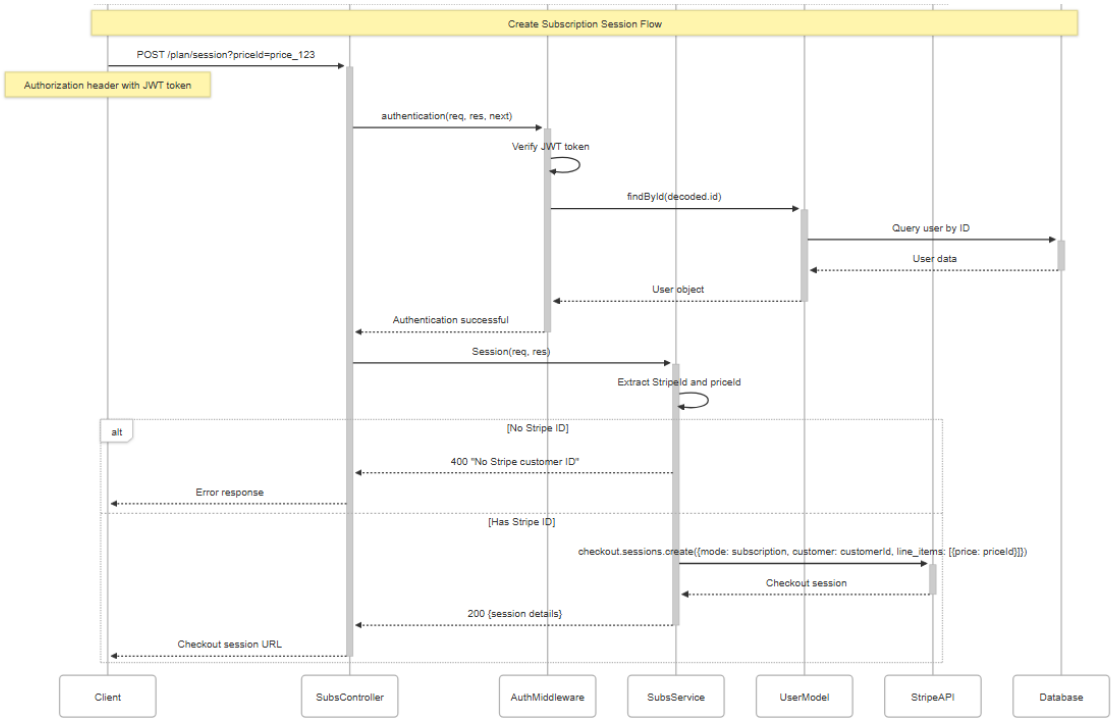
Sign in


Client
AuthController
RegistrationService
UserModel
Database
BCrypt
User Sign In Flow
POST /auth/signin {email, password}
signIn(req, res, next)
findOne({email})
Query user by email
User data or null
User object
alt
[User doesn't exist]
404 "User doesn't exist"
Error response
[User exists]
compare(password, hashedPassword)
Boolean result
alt
[Password incorrect]
404 "Wrong email or password"
Error response
[Password correct]
Generate JWT token (1h expiry)
200 {message, name, token, StripeId, isEmailVerified}
Success response with user data
Client
AuthController
RegistrationService
UserModel
Database
BCrypt

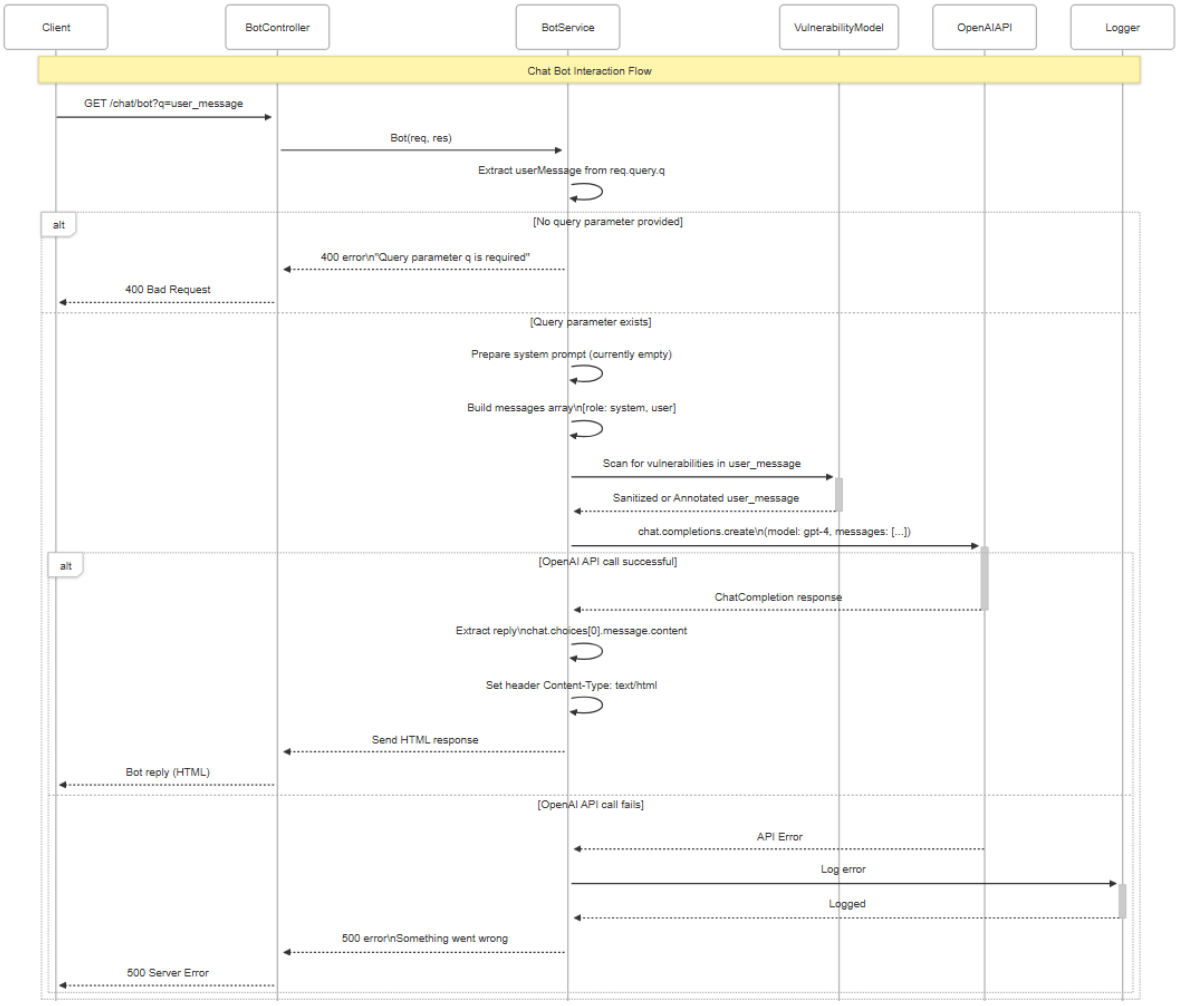
subscrubtion - get subscription



payment session

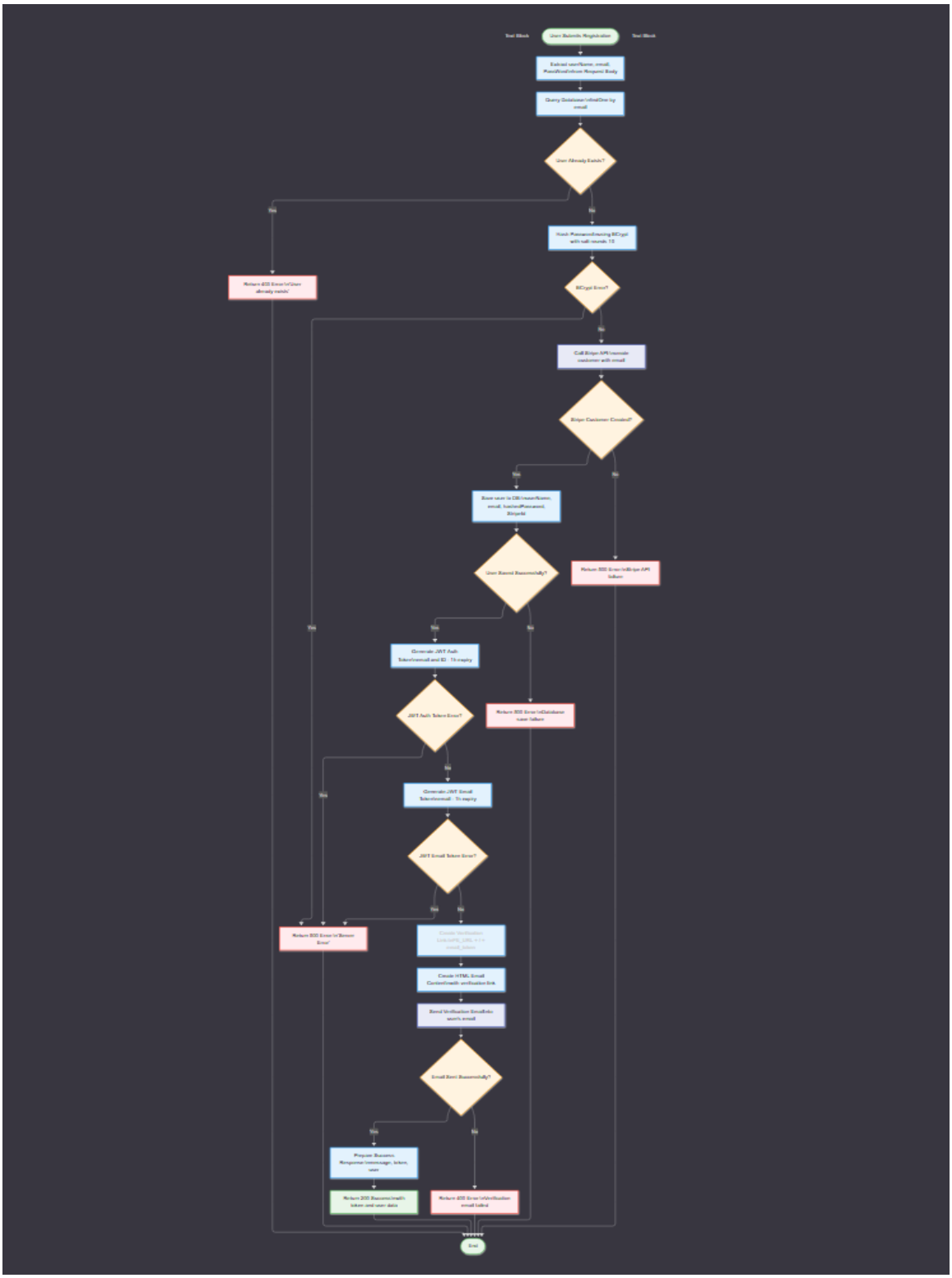


Bot

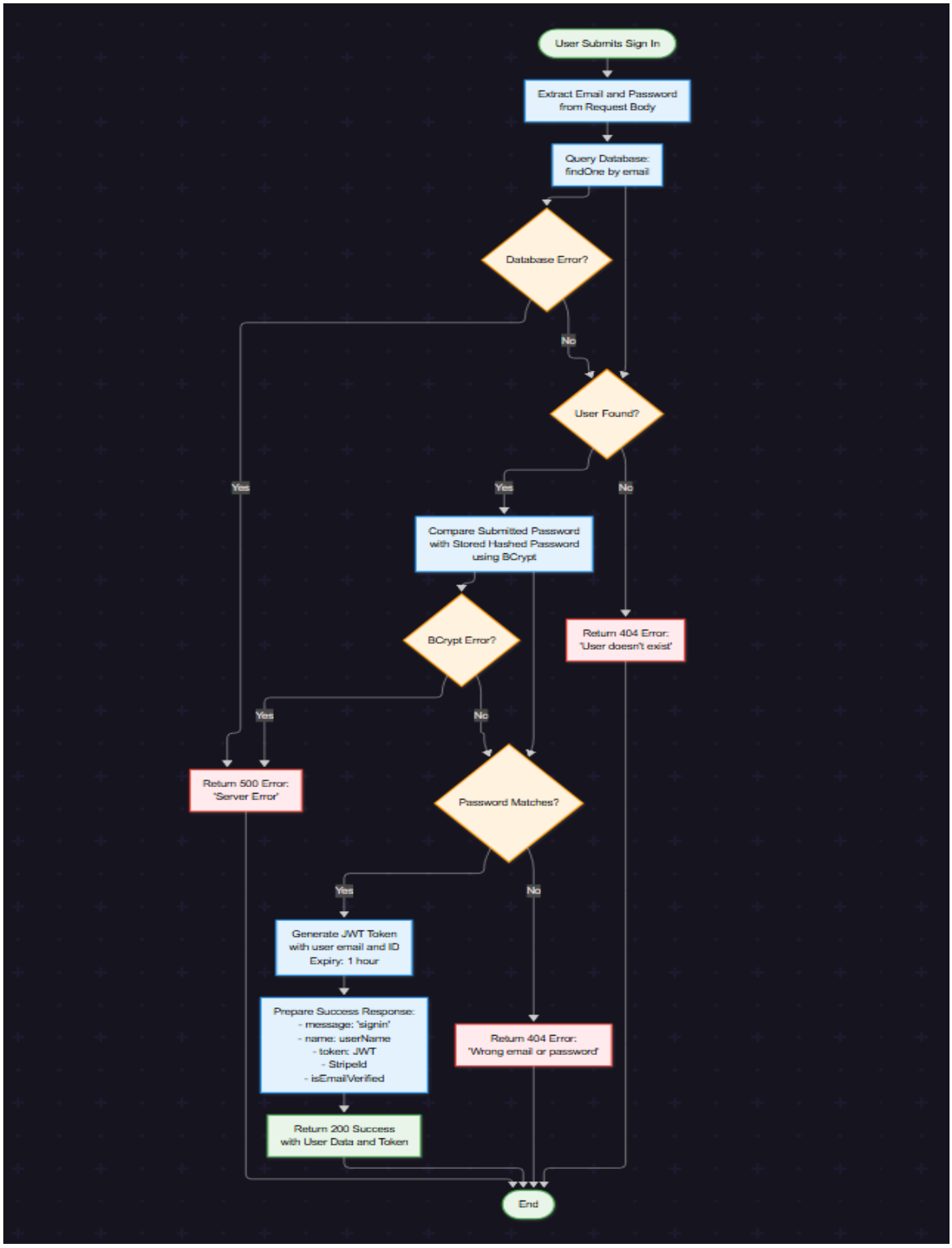

Client
BotController
BotService
VulnerabilityModel
OpenAIAPI
Logger
Chat Bot Interaction Flow
GET /chat/bot?q=user_message
Bot(req, res)
Extract userMessage from req.query.q
alt
[No query parameter provided]
400 error\n"Query parameter q is required"
400 Bad Request
[Query parameter exists]
Prepare system prompt (currently empty)
Build messages array\n[role: system, user]
Scan for vulnerabilities in user_message
Sanitized or Annotated user_message
chat.completions.create\n(model: gpt-4, messages: [...])
alt
[OpenAI API call successful]
ChatCompletion response
Extract reply\nchat.choices[0].message.content
Set header Content-Type: text/html
Send HTML response
Bot reply (HTML)
[OpenAI API call fails]
API Error
Log error
Logged
500 error\nSomething went wrong
500 Server Error

7.6.4 Activity diagrams

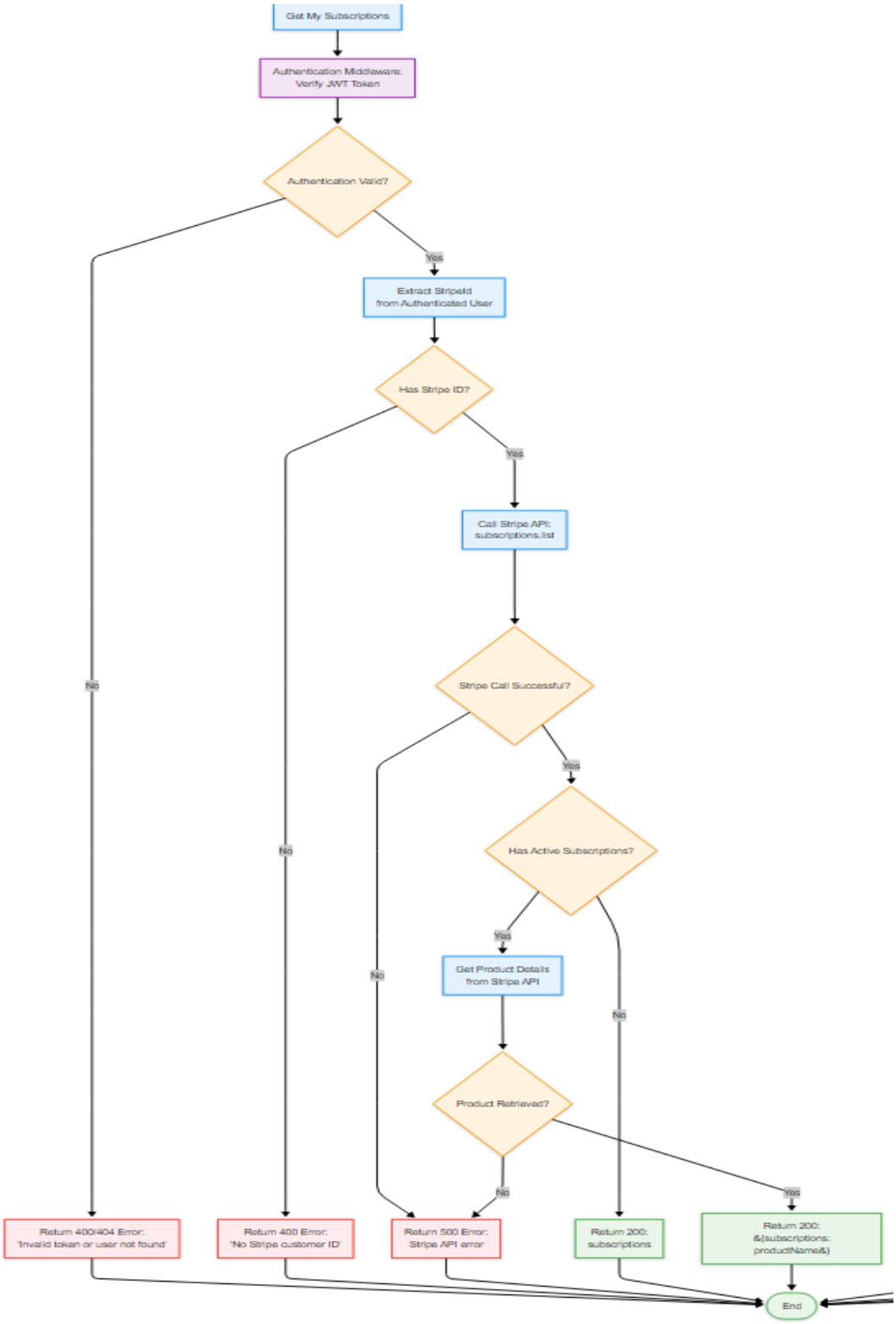
Sign up



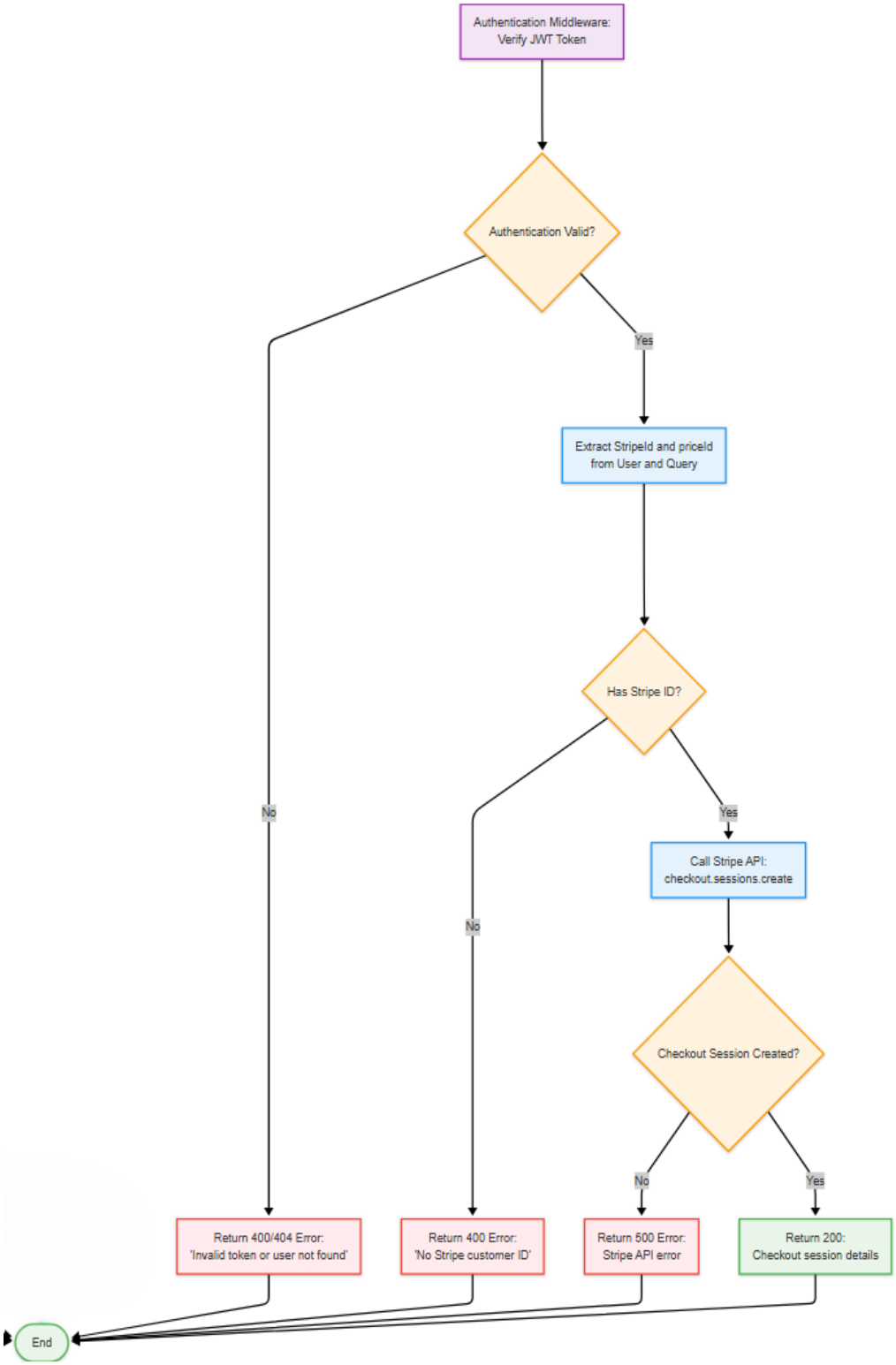
Sign in


User Submits Sign In
Extract Email and Password from Request Body
Query Database: findOne by email
Database Error?
No
User Found?
Yes
Yes
No
Compare Submitted Password with Stored Hashed Password using BCrypt
BCrypt Error?
Return 404 Error: 'User doesn't exist'
Yes
No
Return 500 Error: 'Server Error'
Password Matches?
Yes
No
Generate JWT Token with user email and ID Expiry: 1 hour
Prepare Success Response:
- message: 'signin'
- name: userName
- token: JWT
- StripeId
- isEmailVerified
Return 404 Error: 'Wrong email or password'
Return 200 Success with User Data and Token
End

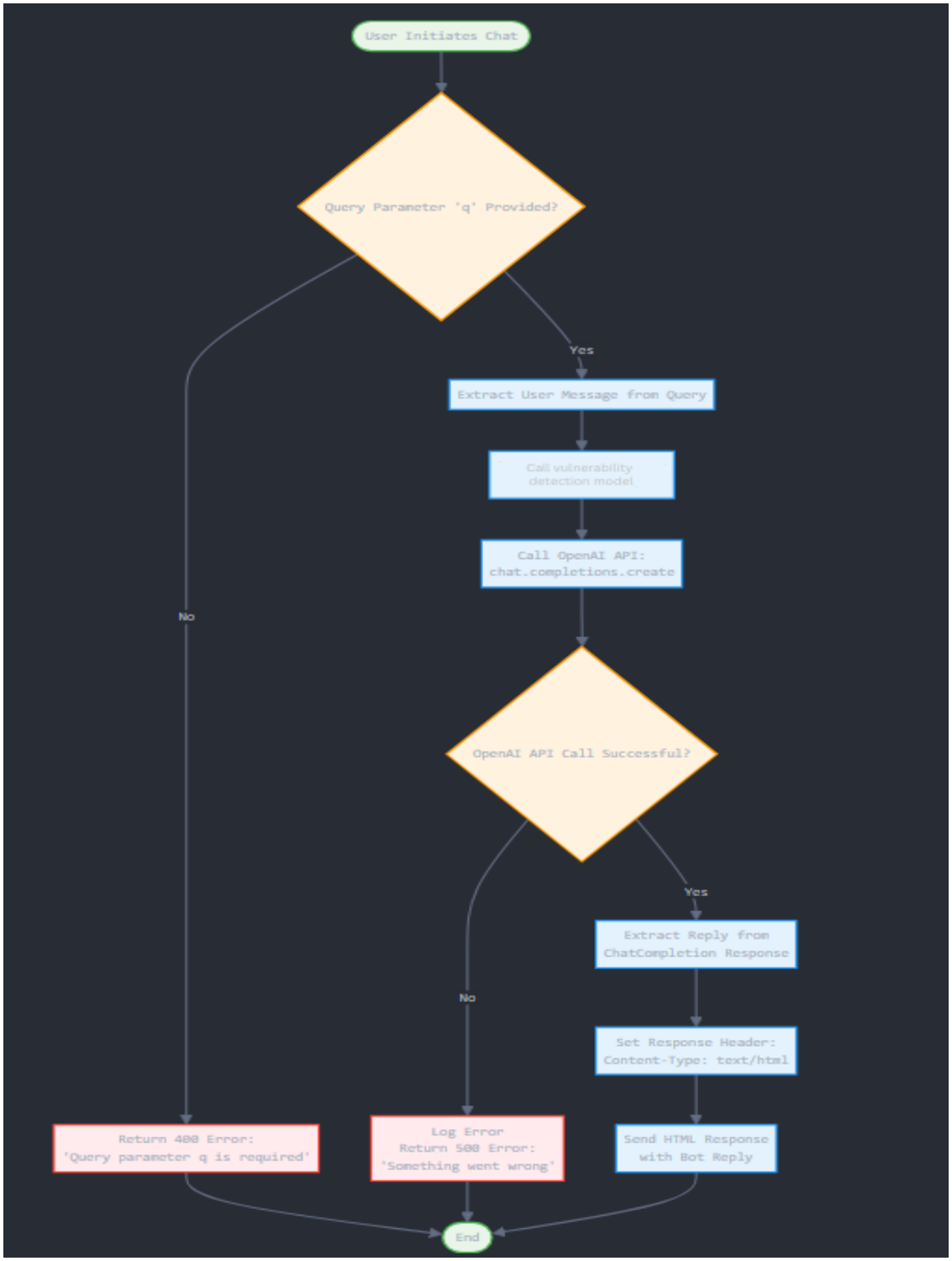
Subscrubtion - get subscription



payment session



Bot


User Initiates Chat
Query Parameter 'q' Provided?
Yes
No
Extract User Message from Query
Call vulnerability detection model
Call OpenAI API: chat.completions.create
OpenAI API Call Successful?
Yes
No
Extract Reply from ChatCompletion Response
Set Response Header: Content-Type: text/html
Return 400 Error: 'Query parameter q is required'
Log Error Return 500 Error: 'Something went wrong'
Send HTML Response with Bot Reply
End

7.7 APIs Documentation:

7.7.1 Authentication Endpoints

7.7.1.1 User Sign up

Creates a new user account and returns a JWT token.

Endpoint: POST Auth /signup

Request Body:

```
{
     "userName": "string",
      "email": "string" ,
     "PassWord": "string",
}
```

Validation Rules:

- Email must be a valid format
- Password must be at least 5 characters

```
•   Success Response (200):
•   {
•     "message": "signup",
•     "token": "string",
•     "user": {
•       "userName": "string",
•       "email": "string",
•       "PassWord": "hashed_string",
•       "StripeId": "string",
•       "isEmailVerified": false,
•       "_id": "string",
•       "createdAt": "timestamp",
•       "updatedAt": "timestamp",  }}
•   Error Response (400): {
•     "errors": [
•       {
•         "msg": "Email already in use"
•       }
•     ],
•   }
```

7.7.1.2 User Login

Authenticates user and returns JWT token.

Endpoint: POST Auth/signin

Request Body:

```
{
  "email": "string",
  "password": "string"
}

Success Response (200):
{
    "message": "signin",
    "name": "String",
    "token": " String ",
    "StripeId": " String ",
    "isEmailVerified": boolean
}
Error Response (400): {
  "errors": [
    {
      "msg": " Invalid credentials"
    }
  ],
}
```

7.7.1.3 Confirm Email

Confirms a user's email address using a token sent via email.

Endpoint: GET /confirm-email/:token

Path Parameter:

token: JWT or unique string sent to the user's email for verification

```
Success Response (200):
{
  "message": "Email confirmed successfully"
}
Error Response (400 or 401):
{
  "message": "Invalid or expired token"
}
```

7.7.2 Bot Endpoints

7.7.2.1 Get Bot Info

Returns bot response based on a query string.

Endpoint: GET/chat /bot

Middleware: authentication

Query Parameters:

q (string, required): The input query to pass to the bot

Example Request:

```
GET /bot?q=Hello
Success Response (200):
{
  "response": "Hi! How can I help you today?"
}
Error Response (400):
{
  "message": "Missing or invalid query parameter 'q'"
}
```

Notes:

- This endpoint require authentication.
- The bot service may return different replies based on LLm and detection model used.

7.7.3 Subscription Endpoints

7.7.3.1 Get Subscriptions

Fetches available subscriptions or user's current plans.

Endpoint: GET /plan/subs

Middleware: authentication

```
Success Response (200):
{
    "subscriptions": "String",
     "active" : boolean,
}
```

7.7.3.2 Create Subscription Session

Creates a session for a new subscription, typically initiating a Stripe checkout session.

Endpoint: POST /session

Middleware: authentication

Query Parameters:

priceId (string, required): The input query to stripe gate way to select plan that want to subscribe

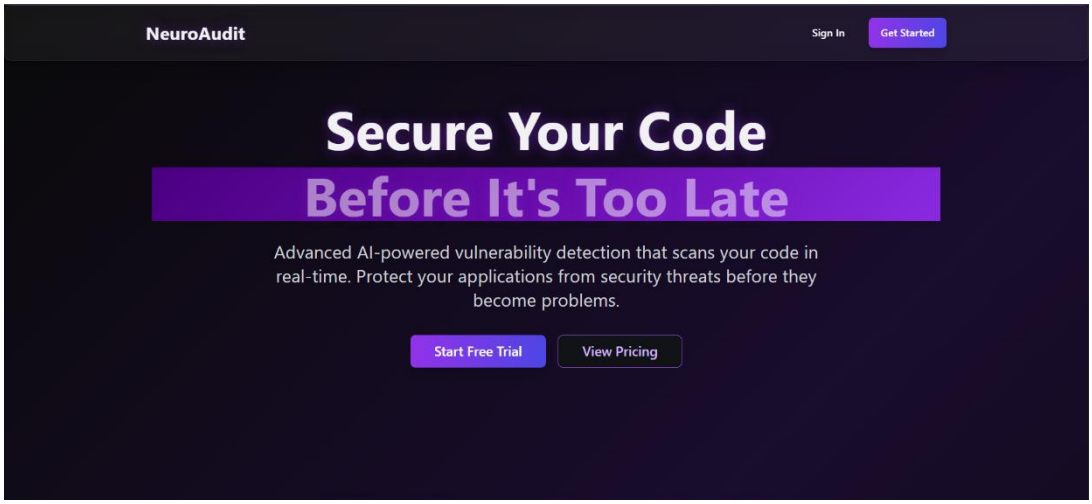
```
Success Response (200):
{
  "sessionId": "cs_test_123456",
  "url": "https://checkout.stripe.com/..."
}
```

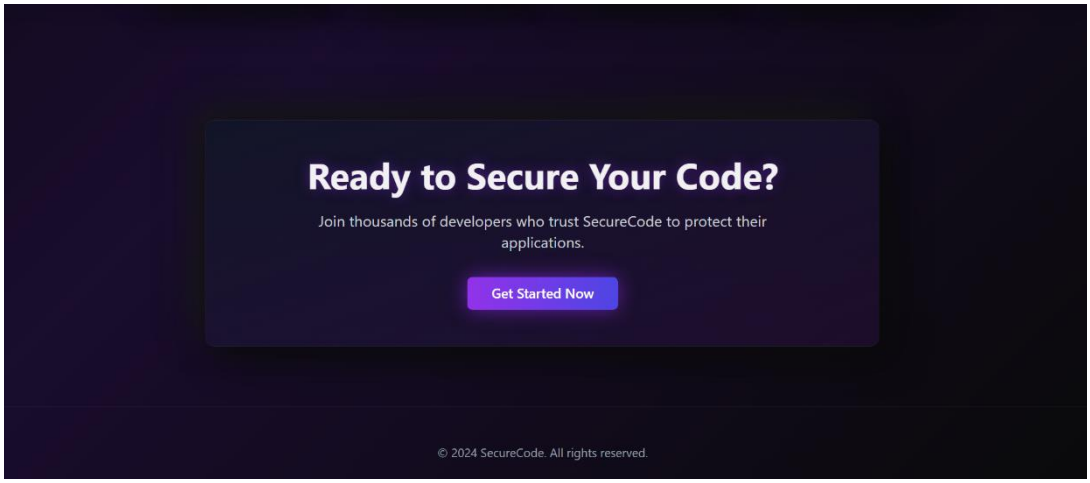
Notes:

- This endpoint require authentication.

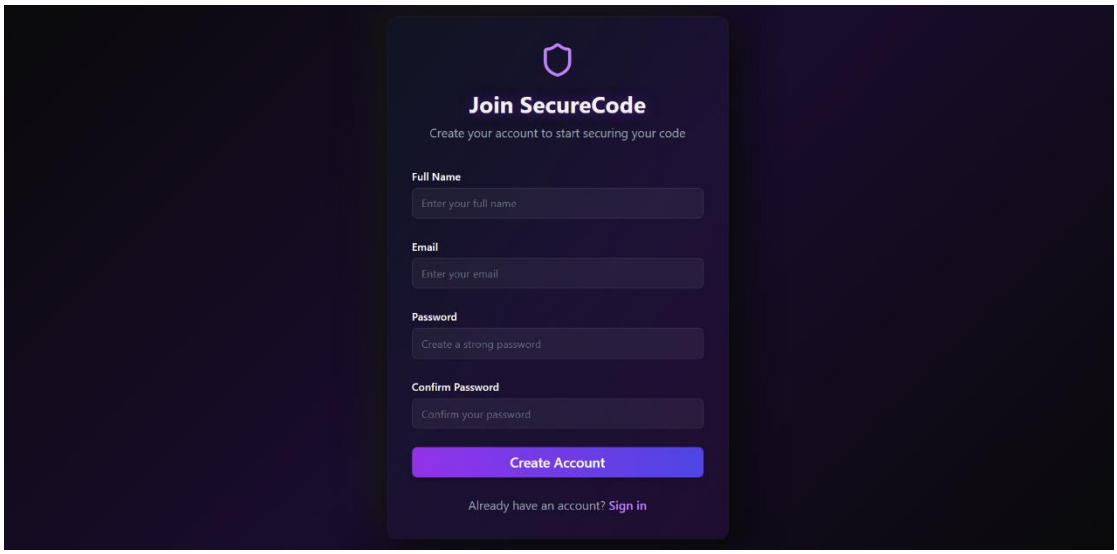
7.8 System UI

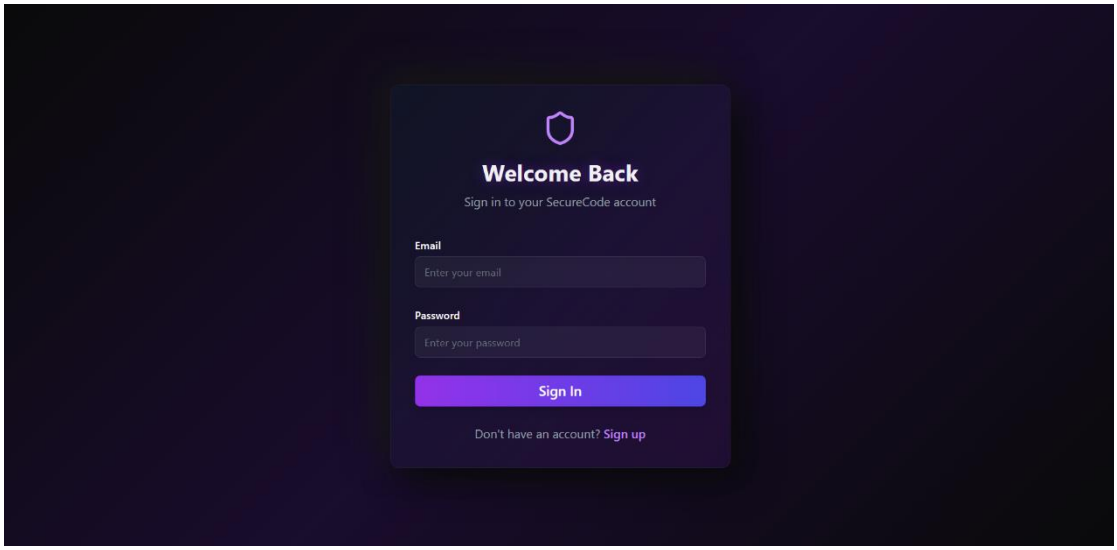
7.8.1 Home Page

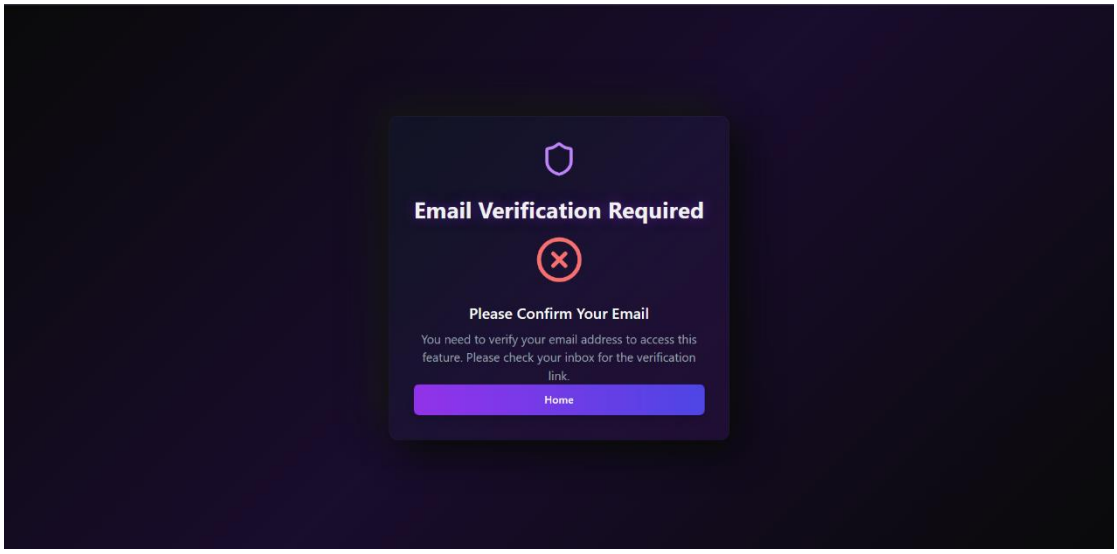




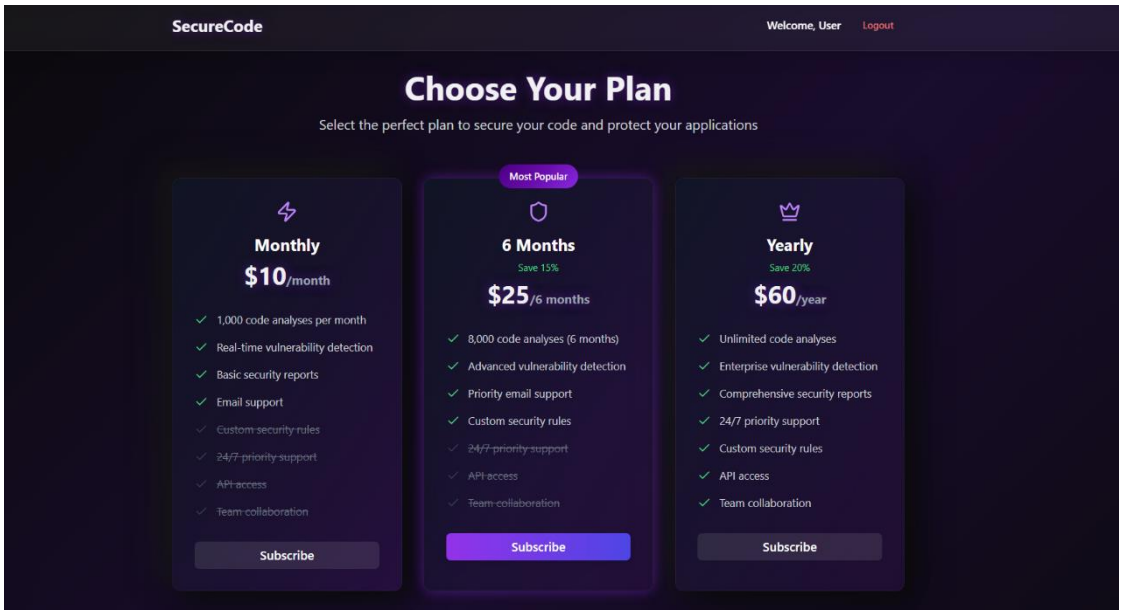
7.8.2 Sign Up & Sign In Page

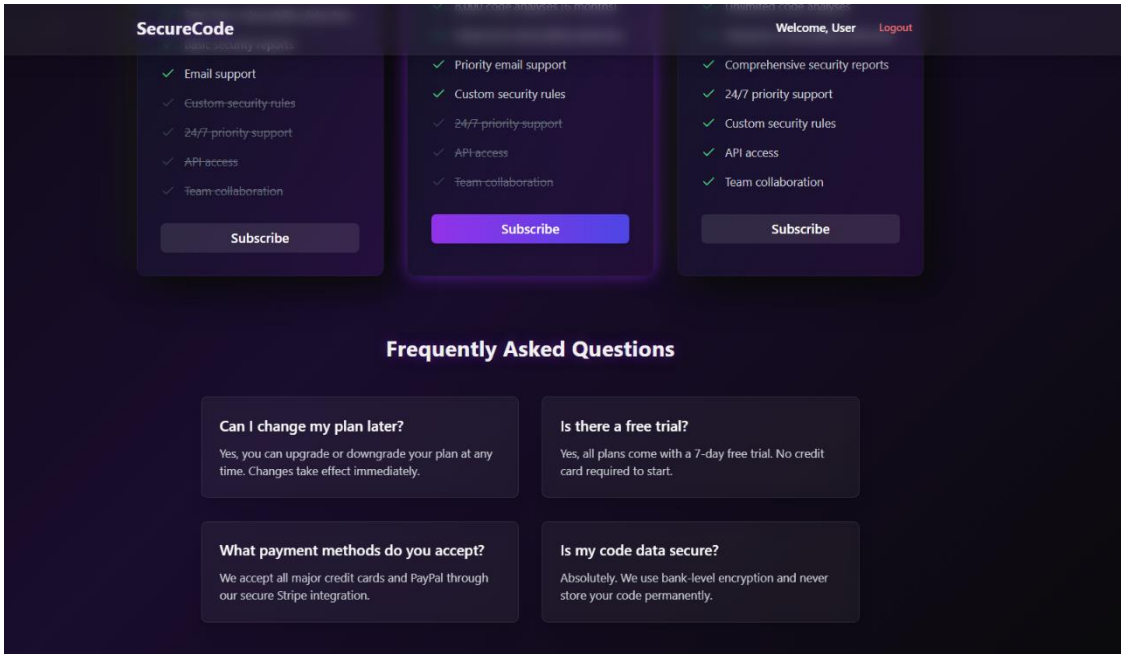



Welcome Back
Sign in to your SecureCode account
Email
Enter your email
Password
Enter your password
Sign In
Don't have an account? Sign up

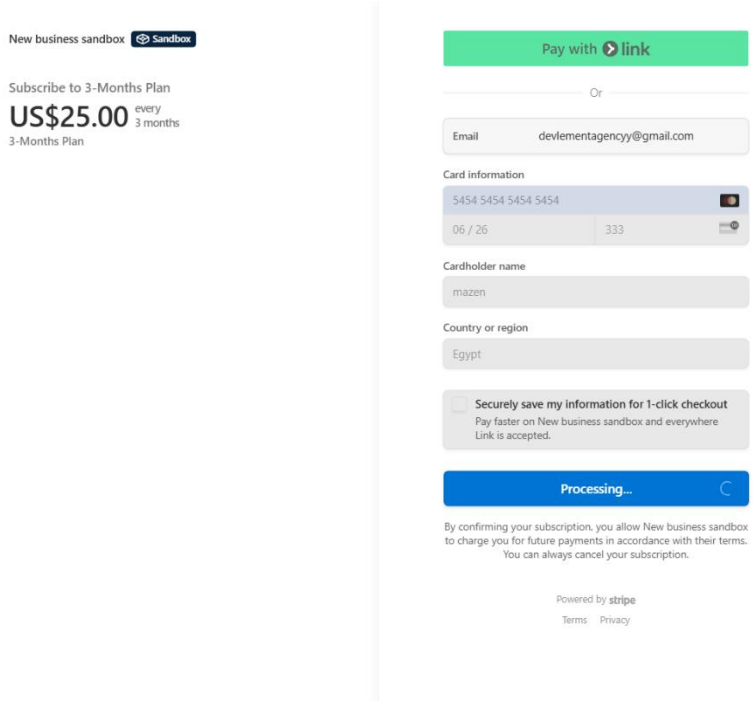

Email Verification Required
Please Confirm Your Email
You need to verify your email address to access this feature. Please check your inbox for the verification link.
Home

7.8.3 Subscribtion Plans

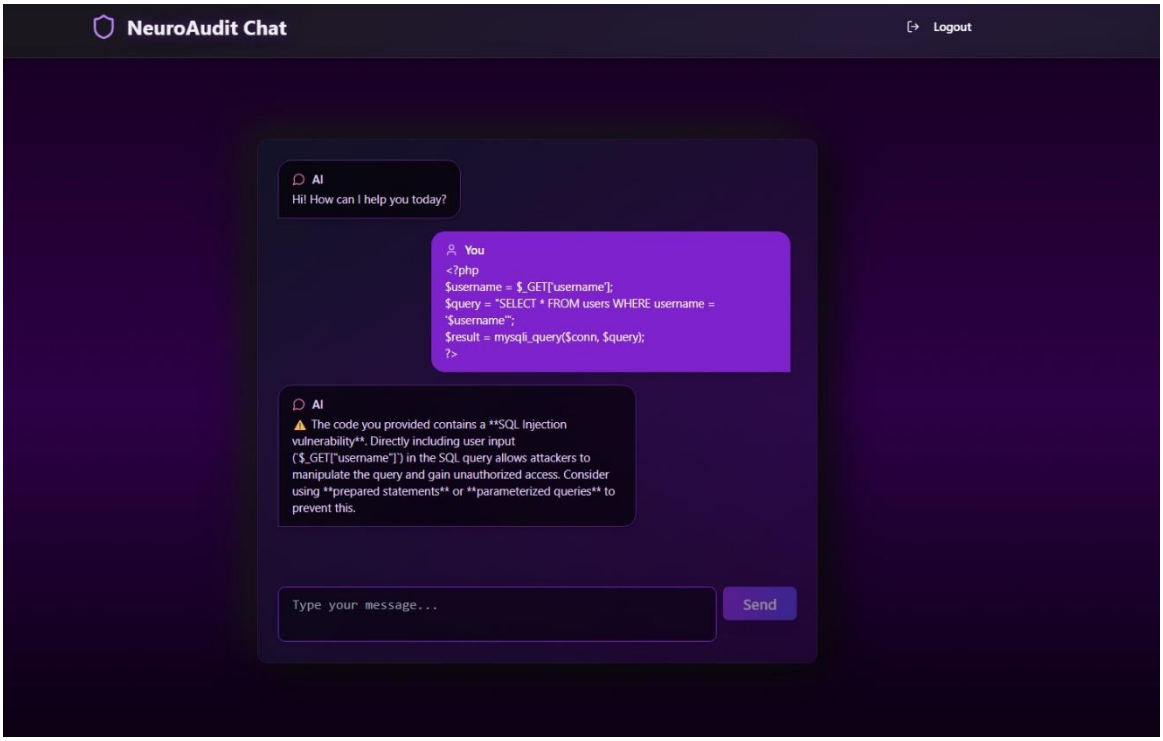




7.8.4 Payment Gateway



7.8.5 Chatbot



8 Chapter 8 – Research Paper

Abstract:

Security weaknesses in web applications frequently stem from vulnerabilities in PHP code, particularly related to SQL Injection (SQLi) and Cross-Site Scripting (XSS). As development practices increasingly incorporate Continuous Integration and Continuous Deployment (CI/CD), there has been a notable rise in the necessity for automated tools that can identify these issues. These tools not only enhance application security but also minimize the substantial time and effort traditionally spent on manual code inspections, which are typically resource-intensive and costly. This document introduces NeuroAudit, a proficient framework driven by neural networks designed to independently evaluate PHP source code to uncover potential SQL Injection (SQLi) and Cross-Site Scripting (XSS) vulnerabilities. The model performs multi-class classification to determine if a segment of code is secure or vulnerable, and if identified as vulnerable, it identifies the specific type of vulnerability. The detection begins with converting the code into an Abstract Syntax Tree (AST), which represents both the syntactic structure and semantic elements of the program. This representation is then tokenized and analyzed using a neural network system that combines Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (BiLSTM) networks. This configuration preserves the code sequence while effectively capturing long-distance dependencies. We evaluated the performance of the system using the PHP Test Suite – XSS, SQLi 1.0.0 (2022) supplied by the Software Assurance Reference Database (SARD). NeuroAudit demonstrated impressive performance, achieving precision, recall, and F1-scores of 82%, 98%, and 90% for SQLi; 100%, 97%, and 99% for XSS; and 96% across all metrics for multi-class classification. Additionally, we confirmed the efficacy of the model using real-world examples sourced from GitHub repositories and Exploit DB, thereby validating its dependability and effectiveness in real-world applications.

Introduction:

In modern business contexts, software-driven applications are essential. Significantly, more than 79.2% of websites are built with PHP, including prominent platforms such as Facebook, Slack, WordPress, Wikipedia, and Spotify. Disturbingly, it has been indicated that cyberattacks take place at a rate of every 39 seconds. Notwithstanding improvements in cybersecurity techniques, PHP applications continue to be vulnerable to notable threats, such as SQL Injection (SQLi) and Cross-Site Scripting (XSS). As evidenced by the Common Vulnerabilities and Exposures (CVE) database, the number of reported vulnerabilities rose from 18,323 in 2020 to 29,066 in 2023, and further increased to 40,300 in 2024, illustrating a marked increase over time. From 2015 to 2025, roughly 10,675 SQL injection vulnerabilities and 31,500 cross-site scripting vulnerabilities were detected, emphasizing the pressing requirement for robust vulnerability detection systems. In order to safeguard websites from such attacks, vulnerability detection technologies are commonly categorized into two classifications: dynamic analysis (performed while the program is executing) and static analysis (conducted without running the program). Static analysis is commonly used in web applications to scrutinize control logic and data flow, frequently leveraging statistical information and pattern recognition techniques to detect vulnerabilities. These static detectors are fundamentally divided into two classifications: code similarity-based and pattern-based. Pattern-based methodologies are further categorized as either rule-based or machine/deep learning-based. Deep learning-based techniques generally begin with preprocessing procedures, including code representation and tokenization, during which source code is converted into a vector format. The model subsequently extracts features and categorizes the code as either vulnerable or clean. Nevertheless, the majority of deep learning models utilized in code analysis—such as BERT or CodeBERT—are resource-intensive, which makes them less appropriate for real-time contexts or applications with constrained resources. To tackle this challenge, we utilize TinyBERT, a streamlined model that provides comparable performance while minimizing memory and processing demands. This research aims to enhance vulnerability detection in PHP applications through an efficient and compact deep learning approach that integrates structured code analysis with a lightweight Transformer-based architecture. The primary contributions of this study include the establishment of preprocessing techniques through the creation of Abstract Syntax Trees (ASTs) for representing source code. Additionally, to enhance the capture of context-sensitive code semantics and long-term dependencies, we integrate TinyBERT as an efficient Transformer-based feature extractor. We execute an exhaustive experimental investigation employing the Software Assurance Reference Database (SARD) along with supplementary out-of-sample datasets to evaluate our methodology in comparison to contemporary state-of-the-art (SOTA) techniques. The findings indicate that the method

we propose outperforms current vulnerability detection techniques with respect to precision, recall, and F1-score. The paper is structured as follows: Section II furnishes essential background information and examines the most relevant research in the domain of software vulnerability detection. Section III delineates the research inquiries and offers a comprehensive overview of our proposed model, elaborating on the methodology utilized for vulnerability identification and the improved technique employed to represent vulnerable features. Section IV details the experimental configuration, encompassing the executed experiments, the employed evaluation metrics, and the resultant findings. It further offers a comparative evaluation of our methodology against current tools and techniques. Section V examines further pertinent literature that corroborates the framework of our research. Section VI discusses the constraints inherent in our methodology. Section VII serves to conclude the paper and delineates prospective avenues for subsequent research.

Background and Related Research:

This section elaborates on key ideas related to identifying vulnerabilities, covering Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Abstract Syntax Tree (AST) representation. The section wraps up by presenting a synopsis of significant academic publications.

A. Deep Neural Network (DNN):

This refers to a category of machine learning models characterized by multiple layers of interconnected nodes or neurons, where each layer adjusts its input data through learned weights and nonlinear activation functions. The word 'deep' refers to the multiple hidden layers situated between the input and output layers, which improve the network's capacity to learn complex, hierarchical data representations.

Key Characteristics:

- Hierarchical Feature Learning: Deep neural networks systematically extract more abstract features from raw data as it moves through layers, thus boosting their effectiveness.
- Nonlinear Transformations: Each neuron applies a nonlinear activation function, enabling the network to capture complex, nonlinear relationships in the data.
- End-to-End Learning: Deep Neural Networks are trained on large datasets and a variety of optimization methods, including backpropagation, to improve weights and minimize prediction errors.

B. Convolutional Neural Network (CNN):

This model is a specialized form of deep learning designed to automatically learn hierarchical features from unprocessed data, especially images. CNNs include several layers, such as convolutional layers which use filters to discover local patterns, pooling layers

that reduce spatial dimensions, and fully connected layers that handle final classification tasks. This architecture enables CNNs to independently learn complex data representations, eliminating the necessity for manual feature engineering.

Principal Characteristics:

- Automatic Feature Extraction: Convolutional Neural Networks (CNNs) succeed in capturing features that range from simple to complex.
- Efficiency and Robustness: Convolutional Neural Networks (CNNs) are adept at processing large datasets efficiently and show resilience to variations in input, including noise or changes in the background.

C. Recurrent Neural Network (RNN):

This type of neural network is designed to evaluate sequential data by maintaining a memory of previous inputs through recurrent connections. This structure enables RNNs to grasp temporal relationships and context within sequences, making them particularly useful for tasks where the sequence order matters. RNNs implement weight-sharing across time steps, enabling efficient learning of patterns in sequential data and retaining information from earlier elements in the sequence for later predictions.

Key Characteristics:

- Sequential Memory: Recurrent Neural Networks (RNNs) maintain a memory function that allows them to use historical context for future predictions.
- Weight Sharing: The cyclic structure of the network supports effective learning and generalization across different points within the sequence.
- Limitations: Traditional recurrent neural networks encounter difficulties in retaining information over long sequences due to issues like vanishing gradients, which can limit their ability to model long-term dependencies.

D. Long Short-Term Memory (LSTM):

LSTM is a sophisticated version of recurrent neural networks (RNNs) created to address the shortcomings of traditional RNNs when it comes to learning long-term connections in sequential information. It incorporates gating mechanisms into its cell design, allowing it to effectively hold onto and utilize crucial data over extended input sequences, a capability that traditional RNNs using basic activation functions lack. The design of an LSTM cell consists of input, output, and forget gates along with state memory, which improves the way it models data based on both time and sequence.

Key features:

- Memory cells: LSTM incorporates specialized units known as memory cells that are capable of retaining information over long durations.
- Gating mechanisms: Long Short-Term Memory networks feature input gates, output gates, and forget gates that manage the flow of

information in and out of the memory cells.

- Constant error flow: LSTM makes use of constant error carousels (CECs) to maintain a steady error flow within specific cells, thus aiding in connecting long time gaps.
- Ability to learn long-term dependencies: LSTM showcases the ability to learn how to navigate minimal time intervals exceeding 1,000 distinct time steps.
- Forget gates: LSTM features forget gates that can learn to reset the internal state of memory cells when the stored information is no longer relevant.

E. Abstract Syntax Tree (AST) Representation:

An Abstract Syntax Tree (AST) is a well-organized, tree-like structure that depicts the syntactic framework of source code, showcasing how its various elements relate to one another. ASTs play a vital role in code representation learning, as they encapsulate both the syntax and, to some extent, the semantics of programs, making them widely used in activities like program analysis, classification, and defect prediction.

Key characteristics:

- Hierarchical Organization: Nodes represent elements such as statements, expressions, and control structures.
- Rich Feature Encoding: Abstract Syntax Trees (ASTs) capture both basic and complex features.
- Semantic Information: By illustrating the code structure and relationships, Abstract Syntax Trees (ASTs) assist in preserving semantic information.

F. Related Work:

a. Static Analysis Tools and Vulnerability Scanners

Static source-code analysis has long been applied to PHP vulnerability detection. Tools like RIPS perform tokenization and taint analysis on PHP ASTs to detect sensitive sinks (e.g. SQL queries, echo statements) vulnerable to user input. Earlier tools such as Pixy and PHP-SAT also used lightweight taint analysis (often regex-based) to flag XSS or SQLi patterns, but they handled only single files or simple code, leading to many false positive. Many open-source PHP scanners exist: for example, Progpilot (2018, MIT license) is a static taint analyzer covering SQLi and XSS with configurable sinks and sanitizers, and tools like Semgrep (with OWASP rules) allow custom pattern matching Industry platforms like SonarQube and GitLab SAST include PHP vulnerability rules (OWASP Top 10, CWE checks), and commercial scanners (e.g. Checkmarx, Veracode) support PHP as well. While such static tools are widely used in practice, published evaluations on PHP are limited; they are known to

generate many false positives and require expert tuning to reduce noise.

b. **Dynamic Analysis and Heuristic Methods**
In parallel, dynamic or black-box testing methods are used to find SQLi/XSS in web apps. Security scanners and fuzzers (e.g. SQLMap, OWASP ZAP, Arachni) inject attack payloads against running PHP applications to discover exploitable inputs. These approaches can confirm real vulnerabilities but require a live environment and cannot directly label source code. Heuristic or pattern-based techniques (e.g. PHP lint rules, custom grep checks, web application firewalls) look for common insecure code idioms (like mysql_query($_GET[...) without sanitization). Such methods are simple to implement but have limited coverage and often miss context-dependent flaws. In summary, traditional heuristics can raise obvious alarms quickly but lack the sophisticated data-flow reasoning of full static analysis.

c. **Machine Learning and Deep Learning Approaches**
Recent research has applied machine learning (ML) and AI to PHP vulnerability classification. Early learning-based models treated vulnerability detection as a classification problem. For example, DEKANT (Medeiros et al., 2016) used a Hidden Markov Model on sequences of code tokens (marked tainted/non-tainted) to characterize PHP vulnerabilities. Similarly, the WAP/WEAPONS framework applied taint analysis with logistic regression and other ML classifiers across multiple vulnerability types, yielding high recall on synthetic samples. Building on these, WIRECAML (Kronjee et al., 2018) extracted rich data-flow features (reaching definitions, taints, constants) from PHP code graphs and trained decision trees. On standard test suites, WIRECAML outperformed static tools (RIPS, Pixy) on most metrics (precision, recall, F1). These data-driven models demonstrate that combining static flow analysis with ML can reduce manual rule-crafting and catch complex patterns better than pure heuristics. More recent work uses deep learning to automatically learn code representations. TAP (Fang et al., 2019) tokenizes PHP and performs simple data-flow slicing, then embeds tokens with word2vec and classifies with an LSTM network, substantially outperforming RIPS in multiclass detection. Likewise, VulHunter (Ahmad et al., 2020) generates CFG/DFG slices around sensitive calls, encodes them (token + bytecode features) and uses a Bi-LSTM; it reports higher recall and F1 on SQLi/XSS than RIP, even

discovering new vulnerabilities in real PHP CMS projects. These deep models remove the need for hand-crafted rules and can capture broader contextual patterns, though they typically require large labeled corpora. The latest trend is to use graph neural networks (GNNs) on code graphs. DeepTective (Rabheru 2020) was the first to apply a hybrid GNN (combining GRU and graph convolutions) to PHP; it models the program CFG to predict XSS/SQLi but has limited public metrics. Most notably, VulEye (Lin et al., 2022) constructs a sub-dependence graph (combining control/data dependencies plus token embeddings) around each sink and applies a graph convolutional network with attention. Overall, AI-driven models (RNNs, GNNs) have demonstrated superior accuracy on benchmark datasets compared to traditional static tools, but they often rely on synthetic or project-specific data and can struggle to generalize to entirely new codebases.

METHODOLOGY:

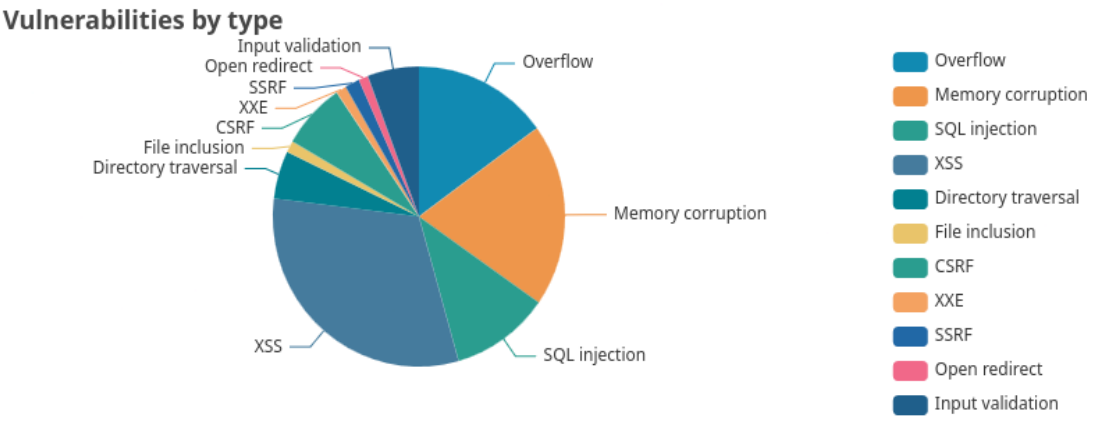
A. Problem Definition

1. Secure Code Review

According to OWASP's guide about Secure Code Review:
"Secure code review is probably the single-most effective technique for identifying security bugs early in the system development lifecycle. When used together with automated and manual penetration testing, code review can significantly increase the cost effectiveness of an application security verification effort."
OWASP defines Secure Code Review as:
"Code review aims to identify security flaws in the application related to its features and design, along with the exact root causes. With the increasing complexity of applications and the advent of new technologies, the traditional way of testing may fail to detect all the security flaws present in the applications. One must understand the code of the application, external components, and configurations to have a better chance of finding the flaws. Such a deep dive into the application code also helps in determining exact mitigation techniques that can be used to avert the security flaws. It is the process of auditing the source code of an application to verify that the proper security and logical controls are present, that they work as intended, and that they have been invoked in the right places. Code review is a way of helping ensure that the application has been developed so as to be "self-defending" in its given environment."

In particular, the following statement highlights the core challenge:
"With the increasing complexity of applications and the advent of new technologies, the traditional way of testing may fail to detect all the security flaws present in the applications."
This clearly emphasizes that secure code review is not a trivial task—it demands significant time, effort, and financial investment. Moreover, due to human limitations, even professional manual code reviews may miss certain flaws.
As a result, the cybersecurity field continuously seeks automation techniques to improve the efficiency and effectiveness of secure code review. While static code analysis tools exist, they are often insufficient on their own. Minor variations in source code structure or logic can lead to false positives or missed vulnerabilities. OWASP also acknowledges this limitation:
"Ideally static code analysis tools would automatically find security flaws with few false positives. That means it should have a high degree of confidence that the bugs that it finds are real flaws. However, this ideal is beyond the state of the art for many types of application security flaws. Thus, such tools frequently serve as aids for an analyst to help them zero in on security relevant portions of code so they can find flaws more efficiently, rather than a tool that finds all flaws automatically."
In light of this, it is evident that cybersecurity reviewers require more elegant and innovative solutions to automate this complex process. No universally concise or reliable method currently exists—this is the gap our research aims to address. We propose an **AI-based Secure Code Reviewer**, developed to assist developers and security analysts in identifying vulnerabilities more effectively and making the digital world safer.

2. **SQL Injection (SQLi)**
   PortSwigger defines SQL Injection as:
   "SQL injection (SQLi) is a web security vulnerability that allows an attacker to interfere with the queries that an application makes to its database. This can allow an attacker to view data that they are not normally able to retrieve. This might include data that belongs to other users, or any other data that the application can access. In many cases, an attacker can modify or delete this data, causing persistent changes to the application's content or behavior. In some situations, an attacker can escalate a SQL injection attack to compromise the underlying



server or other back-end infrastructure. It can also enable them to perform denial-of-service attacks."

This definition clearly highlights the critical nature of SQL injection vulnerabilities and the extensive damage they can cause—from unauthorized data access to complete system compromise.

To underscore the severity and prevalence of SQLi, statistics from CVE Details reveal that over the past ten years, 11,920 SQL injection vulnerabilities have been recorded. In 2024 alone, 2,684 SQLi vulnerabilities were reported. For comparison, Cross-Site Scripting (XSS)—another web vulnerability targeted in our research—recorded 32,430 vulnerabilities, with 7,449 reported in 2024.

The following image provides a visual representation from CVE Details, illustrating the substantial contribution of SQLi and XSS to the overall landscape of web application vulnerabilities.

Now, let us further clarify what SQL Injection (SQLi) is by examining it from a source code perspective, which will help better define our problem domain.

The following two images display PHP source code snippets: the first image shows code vulnerable to SQLi, while the second shows the corrected version.

Focusing on line 4 in the vulnerable code, the program accepts an email input from a typical user. However, at line 5, the developer directly incorporates this input into the SQL query string without any validation or sanitization, implicitly trusting the user-provided data. This approach is risky because if an attacker submits input such as:

abc@example.com'; DROP TABLE users;#

the constructed SQL query at line 5 would become:

SELECT * FROM users WHERE email = 'abc@example.com'; DROP TABLE users;#'

In this query, the attacker cleverly closes the original string literal with a single quote, uses a semicolon to terminate the initial SQL statement, injects a malicious command to drop the users table, and comments out the remaining part with #. This technique exploits the database to perform unintended and destructive actions.

If this explanation is unclear, we will provide additional resources for further understanding.

In the second image, which shows the fixed version of the vulnerable code, the developer employs parameterized queries, also known as prepared statements. In brief, this technique uses placeholders for user-supplied values, compiles the SQL query first, and then safely binds the actual values afterward. This approach prevents user input from being interpreted as SQL syntax, thereby effectively mitigating SQL Injection vulnerabilities.

```php
<?php
$conn = new mysqli("localhost", "username", "password", "databa

$email = $_POST['email'];
$query = "SELECT * FROM users WHERE email = '$email'";

$result = $conn->query($query);
?>
```

```php
<?php
$conn = new mysqli("localhost", "username", "password", "database");

$email = $_POST['email'];
$stmt = $conn->prepare("SELECT * FROM users WHERE email = ?");
$stmt->bind_param("s", $email);
$stmt->execute();

$result = $stmt->get_result();
?>
```

3. **Cross-Site Scripting (XSS)**

PortSwigger defines Cross-Site Scripting (XSS) as:

"Cross-site scripting (also known as XSS) is a web security vulnerability that allows an attacker to compromise the interactions that users have with a vulnerable application. It allows an attacker to circumvent the same origin policy, which is designed to segregate different websites from each other. Cross-site scripting vulnerabilities normally allow an attacker to masquerade as a victim user, to carry out any actions that the user is able to perform, and to access any of the user's data. If the victim user has privileged access within the application, then the attacker might be able to gain full control over all of the application's functionality and data."

Cross-Site Scripting is conceptually similar to SQL Injection and can be considered part of the broader Injection vulnerability family. The key difference lies in the nature of the injected code: instead of injecting SQL syntax to interfere with database queries, XSS involves

injecting malicious JavaScript code.

From a source code perspective, the root cause is the same—the developer fails to properly sanitize or validate user input, assuming it to be trusted. This oversight can lead to critical security impacts on web applications.

Regarding mitigation, unlike SQL Injection which can be effectively addressed with parameterized queries, XSS requires developers to use built-in functions or custom routines to escape or sanitize specific characters in user input. To keep the explanation straightforward, we will not delve into these methods here. Instead, the next two images illustrate an example of vulnerable code susceptible to XSS and its corresponding fixed version.

```php
<?php
// Display user input directly without sanitization
if (isset($_GET['name'])) {
    echo "Hello, " . $_GET['name'];
}
?>
```

```php
<?php
// Escape user input before displaying it to prevent XSS
if (isset($_GET['name'])) {
    echo "Hello, " . htmlspecialchars($_GET['name'], ENT_QUOTES, 'UTF-8');
}
?>
```

Our goal is to develop an effective and reliable AI-based secure code reviewer that automates the identification of the two most common vulnerabilities: SQL Injection (SQLi) and Cross-Site Scripting (XSS). Most importantly, the model aims to detect vulnerabilities based on intrinsic patterns related to SQLi and XSS within the code, regardless of variations such as variable names, values, or other factors.

B. Datasets

The Software Assurance Reference Dataset (SARD) provided by the National Institute of Standards and Technology (NIST) includes two datasets that are highly relevant to our project. While these datasets were originally designed as benchmarks for static code analysis tools, their labeled nature makes them well-suited for machine learning applications as well. By parsing the source code samples and associated labels, we were able to convert each dataset into a structured format (CSV) suitable for training and evaluating our AI model.

It is important to note, however, that the two datasets differ significantly in terms of structure, size, and content. One of the most notable findings from our analysis is that neither dataset fully represents real-world scenarios. This limitation became evident during testing, as both datasets led to results that lacked clear differentiation and generalizability.

To address this, we introduced a third, custom-crafted dataset designed to better simulate real-life vulnerabilities. This dataset includes manually written code

samples as well as vulnerable code snippets extracted from real-world applications available on platforms like GitHub and Exploit DB.

1. **PHP Vulnerability Test Suite (published 27 Oct 2015)**
   This dataset provides a comprehensive collection of 42,212 synthetic PHP test cases, including both vulnerable and corresponding fixed versions, designed to facilitate the study and evaluation of a wide range of security vulnerabilities beyond just XSS and SQL Injection. It is intended for researchers, developers, and security professionals to assess and benchmark the effectiveness of security tools and practices. The dataset includes safe (fixed) and unsafe (vulnerable) PHP code samples, systematically organized by vulnerability type and mapped to their respective CWE (Common Weakness Enumeration) identifiers. Unlike the XSS/SQLi-specific test suite, this dataset covers a broader spectrum of security flaws. These include Insecure Direct Object Reference, classified under CWE-862 (Missing Authorization), and various injection vulnerabilities such as CWE-78 (OS Command Injection), CWE-89 (SQL Injection), CWE-90 (LDAP Injection), CWE-91 (XML Injection), CWE-95 (File Injection), and CWE-98 (PHP Remote File Inclusion). It also addresses issues related to sensitive data exposure, including CWE-311 (Missing Encryption of Sensitive Data) and CWE-327 (Use of a Broken or Risky Cryptographic Algorithm). Additionally, the dataset features examples of security misconfiguration such as CWE-209 (Information Exposure Through an Error Message), improper URL redirects and forwards categorized under CWE-601 (URL Redirection to Untrusted Site), and Cross-Site Scripting vulnerabilities represented by CWE-79. By encompassing such a diverse array of security concerns, this dataset provides a rich resource for training and evaluating AI-based vulnerability detection models.
   One of the main limitations of this dataset is that it contains fewer samples related to our target vulnerabilities—SQL Injection and Cross-Site Scripting—compared to the other dataset, which will be discussed later.

2. **PHP Test Suite - XSS, SQLi 1.0.0 (published 12 May 2022)**
   This dataset comprises a collection of 248,592 synthetic PHP test cases specifically designed to evaluate the effectiveness of static code

analysis tools in detecting SQL Injection (SQLi) and Cross-Site Scripting (XSS) vulnerabilities. While its focus on only two types of vulnerabilities may be seen as a limitation in broader contexts, it aligns well with the scope of our project. The dataset is part of a larger initiative aimed at providing a publicly accessible repository of test cases to support software assurance research and development. The test cases are synthetic, meaning they are programmatically generated to represent a wide variety of vulnerability patterns and sanitization techniques.

3. **SQLI-LABS**
   This is another dataset we discovered on GitHub, consisting of 69 vulnerable PHP source code samples specifically related to SQL Injection. Due to the dataset's imbalance, our team manually created corresponding fixed versions of the 69 samples to achieve a balanced set. However, this dataset was used solely for evaluation purposes and was not included in the training phase.
4. **NeuroAudit's Dataset**
   This is the custom dataset we created, consisting of a mix of vulnerable SQLi PHP source code samples that we developed ourselves, along with real-world examples collected from GitHub and Exploit DB. The final dataset contains a total of 140 samples and was used exclusively for evaluation purposes.

C. **Code Representation**

Undoubtedly, code representation is one of the most critical phases in this project, as it significantly influences both the accuracy and reliability of the AI model. The chosen representation determines which elements of the source code are emphasized, which are omitted, and how various language features—such as data types, syntax, user-defined identifiers, and literal values—are handled. Given its importance, we explored several representation techniques to evaluate their impact on model performance. In the following section, we briefly discuss the different types of code representations we considered and then present a detailed explanation of the one ultimately selected for our model.

1. **Representation Techniques**
   1.1. **Raw Source Code**
   Using raw source code as a representation for AI-based vulnerability detection, without any preprocessing or transformation, is a common but ineffective approach. This method retains the original code, including whitespaces, comments, and formatting, which introduces significant variability. Such variations cause identical code

snippets to appear different to the AI model, hindering its ability to identify consistent patterns for detecting SQL Injection (SQLi) and Cross-Site Scripting (XSS) vulnerabilities. For example, two functionally identical PHP code snippets may differ in whitespace or comments, leading to inconsistent model predictions. While preprocessing techniques, such as removing whitespaces and comments, can mitigate some issues, they offer limited flexibility and fail to capture the semantic structure of the code. In contrast, alternative representations, such as Abstract Syntax Trees (ASTs), provide a more robust and standardized approach, reducing discrepancies and enabling the model to focus on structural and semantic features, as demonstrated in our experiments.

1.2. token_get_all()

The PHP language provides a built-in function, token_get_all(), which parses a given code string into PHP language tokens using the Zend engine's lexical scanner. These tokens represent various components of the PHP language, such as syntax elements, variables, and operators. The token-based representation, illustrated in the subsequent figures showcasing source code and its tokenized form, normalizes the language's syntax and variables into generalized names. To further generalize the code and reduce the vocabulary size, we unified groups of contextually similar tokens into a single token. For example, tokens such as T_VARIABLE (representing variable names) and T_STRING (representing string literals used as identifiers) were combined into a single T_IDENTIFIER token. This unification process streamlined the representation by reducing redundancy and simplifying the token set. Additionally, this approach facilitated the removal of whitespace and special characters, offering a more structured alternative to raw source code. However, the tokenization process exhibited limited performance improvements and lacked sufficient flexibility, requiring numerous manual workarounds that were heavily dependent on the functionality of token_get_all() itself.

```php
<?php
$tokens = token_get_all('<?php echo; ?>');

foreach ($tokens as $token) {
    if (is_array($token)) {
        echo "Line {$token[2]}: ", token_name($token[0]), " ('{$token[1]}')
    }
}
?>
```

```
output.txt

Line 1: T_OPEN_TAG ('<?php ')
Line 1: T_ECHO ('echo')
Line 1: T_WHITESPACE (' ')
Line 1: T_CLOSE_TAG ('?>')
```

1.3. Bytecode

Bytecode representation proved to be one of the most effective approaches in our experiments for processing PHP source code. By converting PHP source code into bytecode using the Zend engine, this method automatically eliminates whitespace, abstracts language syntax, and represents the code as a sequence of opcodes, which function as tokens. This process inherently normalizes variable names and user-defined functions, transforming, for instance, variables like code and name into standardized forms such as var0 and var1, and similarly normalizing function names. Additionally, to further generalize the code and reduce vocabulary size, we unified contextually similar tokens, such as T_VARIABLE and T_STRING, into a single token, T_IDENTIFIER. This normalization, illustrated in the accompanying figures of source code and its tokenized form, aligns closely with the structure of an Abstract Syntax Tree (AST) representation, offering significant automation by eliminating the need for manual preprocessing, as required in the token_get_all() approach. Despite these advantages, the bytecode approach exhibited notable limitations. While it demonstrated strong performance in certain models, its rigidity posed challenges in terms of parsing and flexibility. The bytecode output, generated by the Zend engine in a tabular format, was difficult to parse, necessitating the development of complex Python scripts to extract opcodes and associated data. Furthermore, inconsistencies in the formatting of variable values introduced additional parsing issues, complicating the normalization process. The lack of a consistent mechanism to customize token names or selectively extract components from the bytecode representation limited its adaptability. This trade-off between automation and flexibility ultimately constrained our ability to tailor the normalization process to specific requirements.

1.4. Abstract Syntax Tree (AST)

One of the state-of-the-art techniques for representing source code is the Abstract Syntax Tree (AST), which parses PHP code by traversing it using a depth-first algorithm. This approach abstracts the language's syntax and assigns semantic token names to each entity, such as variables, functions, or user-defined names, regardless of their specific form. The AST distinguishes between built-in and user-defined functions, which is critical for accurate analysis. It also captures both parsed and raw values, providing a comprehensive representation of the code. A key advantage of the AST is its tree-structured JSON format, which significantly enhances flexibility and ease of processing compared to the tabular bytecode representation described earlier. This format allows selective extraction of syntactic components, such as omitting comments or isolating specific tokens relevant to the project, including those specific to HTML syntax embedded in PHP code. For instance, HTML-related tokens can be separated from PHP-specific tokens, enabling targeted analysis. Leveraging this structure, we developed Python scripts to efficiently transform PHP source code into a final AST-based representation, overcoming the parsing challenges and inflexibility encountered with the token_get_all() function and

sample.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple PHP with HTML</title>
</head>
<body>
    <h1>Welcome to My Page</h1>
    <?php
        $username = "Alice";
        $items = ["Book", "Pen", "Notebook"];

        echo "<p>Hello, $username!</p>";
        echo "<ul>";
        foreach ($items as $item) {
            echo "<li>$item</li>";
        }
        echo "</ul>";
    ?>
</body>
</html>
```

bytecode approaches. The resulting representation, depicted in the figures below showcasing PHP source code and its AST, provided a robust and adaptable framework for our analysis, as detailed in subsequent sections.

```
{} ast.json

[
    {
        "nodeType": "Stmt_InlineHTML",
        "attributes": {
            ...
        },
        "value": "<!DOCTYPE html>\n<html>\n<head>\n    <title>Simple PHP with HTML<\/title>\n<\/head>\n<body>\n    <h1>Welcome to My Page<\/h1>\n    "
    },
    {
        "nodeType": "Stmt_Expression",
        "attributes": {
            ...
        },
        "expr": {
            "nodeType": "Expr_Assign",
            "attributes": {
                ...
            },
            "var": {
                "nodeType": "Expr_Variable",
                "attributes": {
                    ...
                },
                "name": "username"
            }
```


approaches, as validated in later sections. The AST approach, implemented using the PHP-Parser library (Nikic, 2022), parses PHP code into a tree-structured JSON format, facilitating

structured traversal and manipulation of code elements.

2. Our Approach

We adopted the Abstract Syntax Tree (AST) representation as the most effective and flexible approach for analyzing PHP source code, particularly for detecting vulnerabilities such as Cross-Site Scripting (XSS) and SQL Injection (SQLi). Our experiments utilized the PHP Test Suite - XSS, SQLi 1.0.0 dataset (published 12 May 2022), which demonstrated superior performance compared to other representations, such as token_get_all() and bytecode

To process the dataset, we converted all PHP source code files into AST JSON-format representations using PHP-Parser. These JSON files were processed with custom Python scripts to extract node names, which served as tokens for subsequent analysis. The processing pipeline included the following steps:

1. **Comment Filtering**: Nodes of type Comment (e.g., inline or block comments) were excluded to focus on functional code elements relevant to vulnerability detection.
2. **Variable Normalization**: For nodes with a name attribute

and type Expr_Variable, the normalize_variable function mapped variable names (e.g., $username, $input) to standardized forms (VAR0, VAR1, etc.), using a dictionary to ensure consistent mappings across the code.

3. **Function Normalization**: For nodes with a name attribute and type Identifier under a Stmt_Function parent node, indicating a user-defined function, the normalize_function function mapped names (e.g., processInput, sanitize) to standardized forms (FUNC0, FUNC1, etc.). For function call nodes (Expr_FuncCall) referencing previously normalized user-defined functions, the same mapping was applied to preserve contextual meaning. Built-in PHP functions (e.g., htmlspecialchars, mysqli_query), crucial for identifying XSS and SQLi vulnerabilities, were preserved without normalization to retain their semantic importance.
4. **Selective Node Extraction**: The scripts enabled flexible selection of nodes, allowing inclusion or exclusion of HTML syntax (represented as Stmt_InlineHTML nodes) and literal values based on project needs.

To assess the impact of HTML syntax and literal values, we generated four AST-based representations:

- **AST with HTML and Values**: Included Stmt_InlineHTML nodes (e.g., HTML markup like Hello) and literal values (e.g., string literals like "Hello").
- **AST with HTML, without Values**: Included Stmt_InlineHTML nodes but replaced literal values with a placeholder (e.g., DUMMY).
- **AST without HTML, with Values**: Excluded Stmt_InlineHTML nodes but retained literal values.
- **AST without HTML, without Values**: Excluded Stmt_InlineHTML nodes and replaced literal values with a placeholder.

Through rigorous experiments on the PHP Test Suite dataset and two out-of-sample datasets (SQLi-LABS and a custom dataset), we determined that the AST representation without HTML syntax and with literal values outperformed the other configurations. This approach preserved critical semantic information (e.g., literal values in SQL queries or XSS payloads) while eliminating irrelevant HTML markup, reducing noise and enhancing model accuracy for vulnerability detection.

Tokenization and Embeddings

To prepare the AST-derived tokens for machine learning, we employed tokenization to convert the sequence of node names and normalized values into numerical representations suitable for model input. Tokenization involves mapping each unique token (e.g., Expr_Variable, VAR0, htmlspecialchars) to a unique integer index, creating a vocabulary that represents the code's structure and content. We used the Keras Tokenizer class (Chollet et al., 2015) from the TensorFlow Keras library to perform this task. The Tokenizer was configured to process the list of tokens generated by the Python script, assigning each token a unique index.

Following tokenization, we transformed the integer sequences into dense vector representations using an embedding layer, a standard technique in natural language processing and code analysis. Embeddings map each token index to a fixed-length vector of real numbers,

tokens.txt

```
Stmt_InlineHTML Stmt_Expression Expr_Assign Expr_Variable VAR1 Scalar_String Alice Stmt_Expression
Expr_Assign Expr_Variable VAR2 Expr_Array ArrayItem Scalar_String Book ArrayItem Scalar_String Pen
ArrayItem Scalar_String Notebook Stmt_Echo Scalar_InterpolatedString InterpolatedStringPart
<p>Hello,  Expr_Variable VAR1 InterpolatedStringPart !</p> Stmt_Echo Scalar_String <ul> Stmt_Foreach Expr_Variable
VAR2 Expr_Variable VAR3 Stmt_Echo Scalar_InterpolatedString InterpolatedStringPart <li> Expr_Variable VAR3 InterpolatedStringPart
</li> Stmt_Echo Scalar_String </ul> Stmt_InlineHTML
```

capturing semantic relationships between tokens based on their co-occurrence in the code. We incorporated a Keras Embedding layer at the start of our neural network architec-

sample.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple PHP with HTML</title>
</head>
<body>
    <h1>Welcome to My Page</h1>
    <?php
        $username = "Alice";
        $items = ["Book", "Pen", "Notebook"];

        echo "<p>Hello, $username!</p>";
        echo "<ul>";
        foreach ($items as $item) {
            echo "<li>$item</li>";
        }
        echo "</ul>";
    ?>
</body>
</html>
```

ture to generate these embeddings. The Embedding layer was initialized with a predefined vocabulary size (derived from the Tokenizer) and an embedding dimension, producing dense vectors that were fed into subsequent layers for vulnerability classification. This approach allowed the model to learn contextual representations of PHP code elements, such as distinguishing between built-in functions like htmlspecialchars (indicative of XSS sanitization) and user-defined functions.

The Keras Tokenizer and Embedding layer provided a robust pipeline for converting AST tokens into a format suitable for deep learning models. The tokenization step ensured a consistent

numerical representation, while the embedding layer enabled the model to capture semantic patterns critical for detecting XSS and SQLi vulnerabilities. The results of this pipeline, applied to the AST-without-HTML-with-values representation, are illustrated in the accompanying figures, which depict a sample PHP source code with embedded HTML and its corresponding AST.

D. Model Architecture

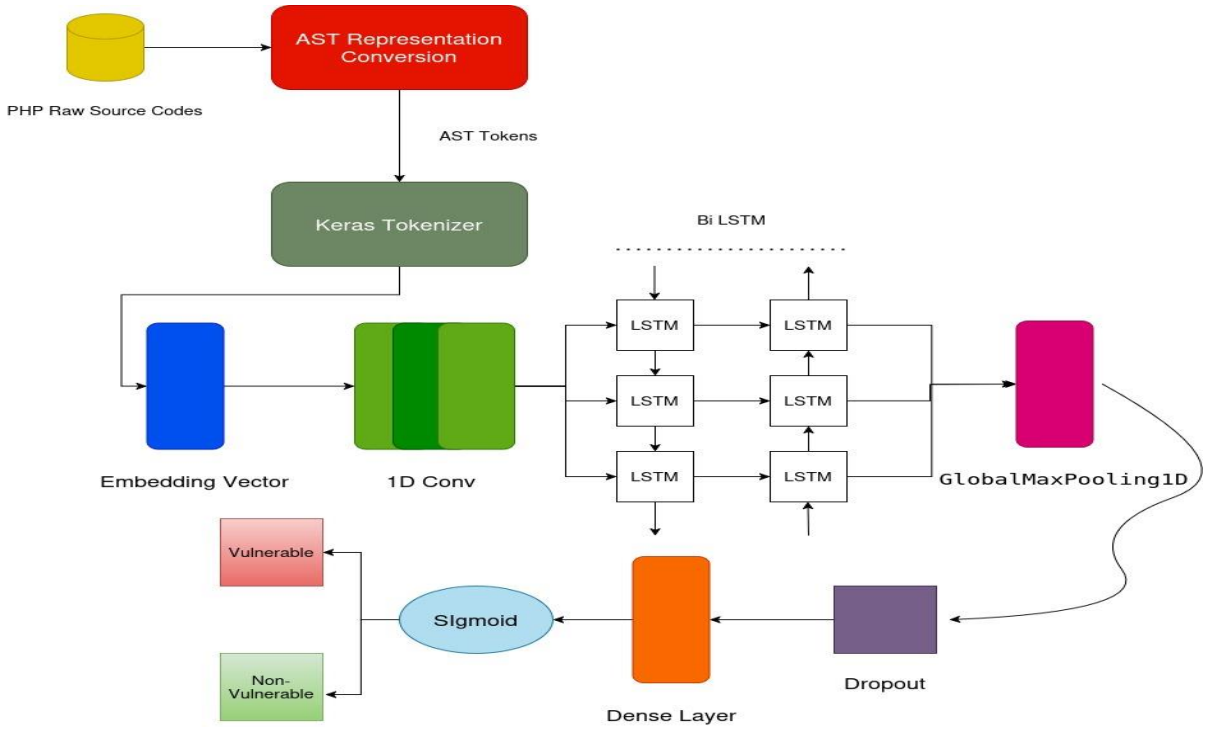
After obtaining the embedding vectors from the embedding layer, we feed them into our proposed model architecture, which is a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) layers. The design of this architecture aims to capture both local syntactic patterns and long-range semantic dependencies within the source code.

The architecture consists of the following layers:

- **Conv1D Layer**:
  We begin with a one-dimensional convolutional layer with 64 filters, a kernel size of 3, 'same' padding, and ReLU activation. This layer is responsible for detecting local patterns in the input sequences—such as common vulnerability patterns or recurring syntax structures—by applying sliding filters over the embedding sequences.
- **Bidirectional LSTM Layer**:
  Next, we use a Bidirectional LSTM with 64 units and return_sequences=True. This layer processes the sequence in both forward and backward directions to learn context from both past and future tokens. This is especially beneficial for source code, where dependencies can exist in both directions, such as variable declarations affecting later and earlier statements.
- **GlobalMaxPooling1D**:
  After the recurrent layer, we apply Global Max Pooling to reduce the sequence of vectors to a fixed-length representation by selecting the maximum value across each feature dimension. This layer helps retain the most salient features extracted by previous layers while reducing dimensionality, making the model less prone to overfitting.
- **Dropout Layer**:
  A dropout layer with a rate of 0.5 is applied for regularization. This prevents overfitting by randomly deactivating 50% of the neurons during training, encouraging the model to generalize better on unseen data.
- **Dense Layer (64 units)**:
  A fully connected layer with 64 units and ReLU activation is used to further process the pooled features and introduce non-linearity. This layer acts as

a bridge between the high-level representations and the final classification layer.

- **Output Dense Layer**:
  The final layer is a dense layer with the number of output units equal to the number of classes (i.e., len(y_test[0])), and a soft-max activation function. This layer produces a probability distribution over the output classes, allowing the model to classify the input code as either vulnerable or non-vulnerable to SQL Injection or XSS.

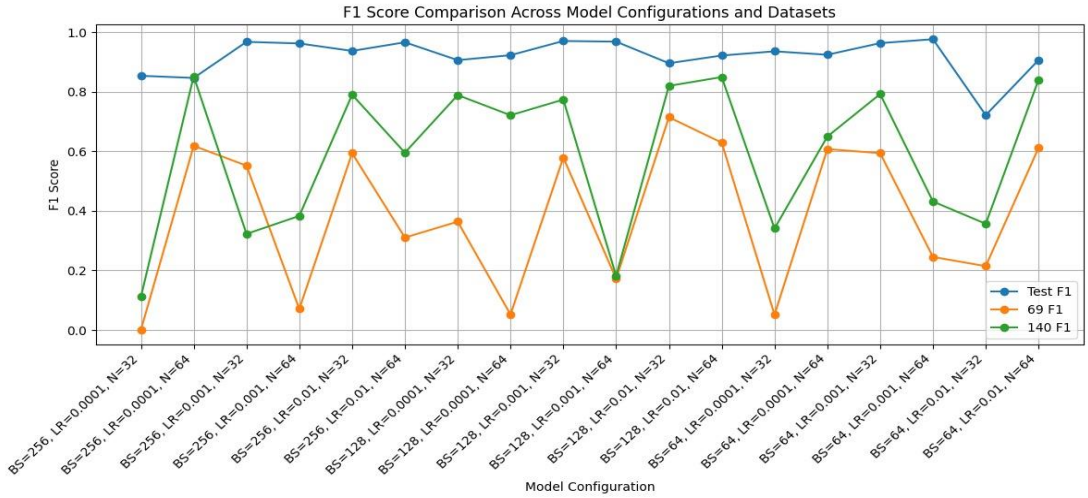


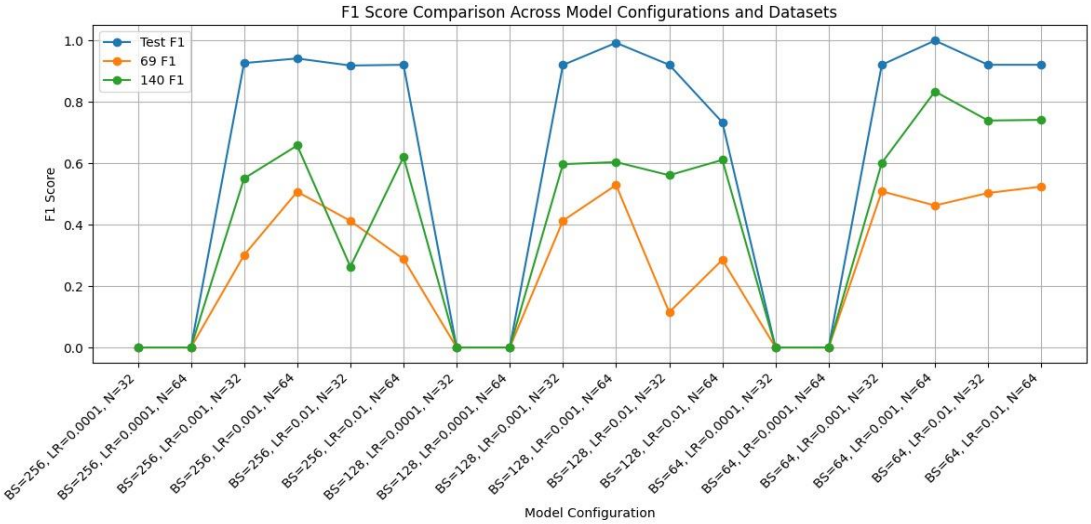
This architecture was selected after iterative experimentation and comparison with alternative designs. The combination of convolutional and recurrent layers allows the model to exploit both structural and sequential information present in the source code representations, ultimately improving its ability to detect complex vulnerability patterns.

Enviromental and Experimental Work:

Our team conducted a series of experiments involving different techniques throughout the project,
including variations in datasets, Abstract Syntax Tree (AST) representations, and model architectures. Initially, we used performance on SQL Injection (SQLi) vulnerability detection as the primary evaluation criterion to select the most suitable dataset, representation method, and model design. Based on the results, we identified the optimal configuration and proceeded to develop two additional models: a binary classification model for detecting Cross-Site Scripting (XSS) vulnerabilities, and a multi-class classification model capable of identifying both SQLi and XSS vulnerabilities.

A. Datasets





We evaluated two primary datasets from NIST for model training: one published in 2015 and another in 2022. A detailed comparison of this project, we focused exclusively on the SQL Injection (SQLi) samples from both datasets, as SQLi was our initial target vulnerability. We assessed the performance of these datasets using various model configurations, testing them on their respective test sets and two external validation datasets—SQLI-LABS and our custom dataset. As shown in the own test data in terms of accuracy and F1 score. However, the 2022 dataset consistently achieved better results on the independent validation datasets. Based on these findings, we selected the 2022 dataset as the basis for further development.these datasets is provided in the Datasets section. For the purposes of figures below, both datasets demonstrated strong performance on their

B. AST Representation

As previously discussed, we explored four approaches to representing the Abstract Syntax Tree (AST) of our PHP source code. After extensive experimentation, we observed that including HTML content in the representation led to poor performance and significantly increased the token count and

```
              precision    recall  f1-score   support

           0       0.99      0.95      0.97     21918
           1       0.82      0.98      0.90      5664

    accuracy                           0.95     27582
   macro avg       0.91      0.96      0.93     27582
weighted avg       0.96      0.95      0.95     27582

ROC AUC: 0.9846075130829187
Confusion Matrix:
 [[20738  1180]
 [  115  5549]]
```

vocabulary size. Consequently, we focused on the variants that excluded HTML. Among these, we evaluated

whether to retain literal values or replace them with a placeholder such as DUMMY. The results showed that, for SQLi vulnerability detection, retaining the actual values consistently yielded better performance across all evaluation metrics. A summary of these results is presented in the figure below.

C. Model Architecture

After extensive experimentation and selecting the 2022 dataset as our primary training source, we conducted a grid search focused on SQLi vulnerability samples to identify the optimal hyperparameters. The configuration that yielded the best performance—based on results from both the test set and the two validation datasets—used a batch size of 128, a learning rate of 0.01, and 64 neurons in the BiLSTM layer.

	Batch Size	LR	Neurons	Test Acc	Test F1	Test Loss	69 Acc	69 F1	69 Loss	140 Acc	140 F1	140 Loss
0	256	0.0001	32	0.942716	0.853351	0.148343	0.437956	0.000000	1.728505	0.314286	0.111111	1.328062
1	256	0.0001	64	0.935900	0.846181	0.156010	0.540146	0.618182	0.939221	0.771429	0.851852	0.557010
2	256	0.0010	32	0.986585	0.967297	0.043154	0.525547	0.551724	1.829781	0.400000	0.322581	2.314865
3	256	0.0010	64	0.984809	0.961968	0.035563	0.423358	0.070588	2.420525	0.378571	0.382979	1.948853
4	256	0.0100	32	0.974331	0.936820	0.064976	0.540146	0.593548	4.652956	0.707143	0.789744	2.506711
5	256	0.0100	64	0.985715	0.965840	0.033452	0.416058	0.310345	4.332488	0.514286	0.595238	2.884881
6	128	0.0001	32	0.962367	0.905893	0.104696	0.540146	0.363636	0.913788	0.735714	0.788571	0.487889
7	128	0.0001	64	0.968385	0.922503	0.074533	0.467153	0.051948	2.600781	0.657143	0.720930	1.304274
8	128	0.0010	32	0.987492	0.970174	0.027057	0.532847	0.578947	2.341009	0.707143	0.773481	1.378687
9	128	0.0010	64	0.986875	0.967987	0.030911	0.437956	0.172043	2.623924	0.292857	0.181818	2.475501
10	128	0.0100	32	0.953049	0.895506	0.113884	0.708029	0.714286	2.664279	0.735714	0.819512	1.712639
11	128	0.0100	64	0.965811	0.921436	0.069802	0.569343	0.628931	3.597943	0.771429	0.849057	1.752998
12	64	0.0001	32	0.974875	0.935469	0.068498	0.459854	0.051282	1.712540	0.364286	0.340741	1.753014
13	64	0.0001	64	0.967950	0.923977	0.083507	0.613139	0.607407	1.129567	0.614286	0.649351	0.816362
14	64	0.0010	32	0.985135	0.962976	0.035252	0.540146	0.593548	1.867483	0.700000	0.792079	1.800706
15	64	0.0010	64	0.990247	0.976138	0.020349	0.416058	0.245283	2.219247	0.378571	0.431373	1.901664
16	64	0.0100	32	0.908056	0.721257	0.158676	0.518248	0.214286	2.288344	0.407143	0.356589	1.373133
17	64	0.0100	64	0.959684	0.906208	0.094621	0.554745	0.611465	3.726911	0.750000	0.840183	0.993856

Results and Discussion:

Following the completion of our experiments using the dataset published in 2022—and after demonstrating its superiority over the 2015 dataset using out-of-sample evaluation on SQLiLABS and our custom dataset—we proceeded with training our models using the optimal combination of model architecture, hyperparameters, and AST representation. We developed two additional models: a binary classifier for XSS detection and a multi-class model for detecting both SQLi and XSS vulnerabilities. All three models demonstrated strong performance on both in-sample test data and out-of-sample real-world evaluation datasets. The results for the three models are summarized below.

Overall, our model demonstrated superior performance on the dataset published in 2022 compared to the 2015 dataset, particularly when evaluated against newly introduced validation datasets that reflect real-world scenarios. In future work, we aim to enhance the project by expanding the scope of supported vulnerabilities, incorporating additional programming languages, and integrating a

vulnerability locator capable of identifying the exact line of code responsible for each detected issue.

	accuracy	f1 score	recall	precision
SQLi	0.95	0.90	0.98	0.82
XSS	1.00	0.99	0.97	1.00

SQLi & XSS	f1 score	recall	precision
macro avg	0.96	0.96	0.96
weighted avg	0.98	0.98	0.98

Limitations:

The constraints encountered by this research that warrant consideration for future endeavors include:

- NeuroAudit is a PHP code reviewer that examined the PHP Test Suite – XSS, SQLi 1.0.0 (2022) dataset, which, however, does not reflect real-world scenarios. Most files were limited to fewer than 50 lines and contained only fragments of vulnerabilities, failing to encompass all potential cases. Furthermore, the conversion of the dataset to Abstract Syntax Trees (ASTs) revealed numerous duplicated files, likely resulting from prior data augmentation. As a result, we sought to identify or construct alternative examples. However, due to time constraints, we collected approximately 200 files designed to represent a range of real-world scenarios.
- In addition, limited computational resources restricted our ability to explore advanced methods such as bytecode analysis, as our hardware was unable to process more than 4,000 samples. Nonetheless, bytecode remains a promising direction for future research due to its potential to provide deeper semantic insights.

Conclusion and future work:

In conclusion, NeuroAudit offers a model that benefits software applications and all users of the PHP language, promoting the creation of vulnerability-free code and thus improving their cybersecurity. Utilizing deep learning techniques and a hierarchical tree structure, the model demonstrates efficacy in identifying PHP vulnerabilities, SQL injection (SQLi), and cross-site scripting (XSS). The NeuroAudit model utilizes Abstract Syntax Tree (AST) representation to develop a structured tree that abstracts the syntax of the language and associates semantic token names. Furthermore, it encompasses both interpreted and unprocessed values, thus providing a comprehensive representation of the code. Subsequently, we tokenize the output of the Abstract Syntax Tree

(AST) to enable its incorporation into a hybrid deep learning framework that includes Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory networks (BiLSTM). As a result, the sequencing of the code can be maintained through the application of BiLSTM, which further aids in the administration of long-term dependencies. The empirical results derived from the benchmark dataset illustrate the considerable capacity of the proposed model to enhance the performance and precision of identifying software vulnerabilities in PHP source code files. NeuroAudit sets itself apart from current research in this field, especially regarding vulnerability detection.

Addressing the previously identified limitations, subsequent research will emphasize the expansion of NeuroAudit to include a wider range of real-world samples and varied scenarios. Furthermore, we are committed to broadening our expertise to include additional programming languages, an objective that is theoretically attainable through the implementation of Abstract Syntax Tree (AST) representation; nevertheless, we plan to undertake experimentation in this domain. Additionally, this study will examine further and varied vulnerabilities to improve the generalization abilities of NeuroAudit, signifying an area of emphasis for future research.

Ahead. arXiv preprint arXiv:2304.05678, 2023.

9 Chapter 9 - Conclusions and Future Work

9.1 Summary

In conclusion, NeuroAudit offers a model that benefits software applications and all users of the PHP language, promoting the creation of vulnerability-free code and thus improving their cybersecurity. Utilizing deep learning techniques and a hierarchical tree structure, the model demonstrates efficacy in identifying PHP vulnerabilities, SQL injection (SQLi), and cross-site scripting (XSS). The NeuroAudit model utilizes Abstract Syntax Tree (AST) representation to develop a structured tree that abstracts the syntax of the language and associates semantic token names. Furthermore, it encompasses both interpreted and unprocessed values, thus providing a comprehensive representation of the code. Subsequently, we tokenize the output of the Abstract Syntax Tree (AST) to enable its incorporation into a hybrid deep learning framework that includes Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory networks (BiLSTM). As a result, the sequencing of the code can be maintained through the application of BiLSTM, which further aids in the administration of long-term dependencies. The empirical results derived from the benchmark dataset illustrate the considerable capacity of the proposed model to enhance the performance and precision of identifying software vulnerabilities in PHP source code files. NeuroAudit sets itself apart from current research in this field, especially regarding vulnerability detection.

9.2 Future Work

To address the current limitations and enhance the capabilities of *NeuroAudit*, future work will focus on the following directions:

- **Expand Real-World Dataset Coverage**:
  Increase the diversity and volume of training and evaluation data by including more real-world code snippets. This will improve generalization and robustness across different coding styles, obfuscation techniques, and use cases.
- **Support Multiple Programming Languages**:
  Extend the system to analyze additional languages beyond PHP. This can be

approached by using language-agnostic representations such as **Abstract Syntax Trees (ASTs)** or **intermediate representations**, enabling model transferability and cross-language vulnerability detection.

- **Explore More Vulnerability Types**:
  Move beyond SQLi and XSS to include vulnerabilities such as Command Injection, Path Traversal, and Insecure Deserialization. This would strengthen the model's general applicability to real-world secure code auditing.
- **Build a Feedback Loop for Continuous Learning**:
  Develop a mechanism for integrating real-world feedback from developers or security analysts into model retraining, making the system adaptive and continuously improving.

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الملخص (Abstract in Arabic):

- تنشأ العديد من نقاط الضعف الأمنية في تطبيقات الويب من ثغرات في شفرةPHP ، وخاصة تلك المتعلقة بحقن أوامر SQL (SQL Injection - SQLi)وتنفيذ التعليمات البرمجية عبر المواقع .(Cross-Site Scripting - XSS) ومع تطور ممارسات التطوير نحو الاعتماد المتزايد على التكامل المستمر والنشر المستمر(CI/CD) ، برزت الحاجة المتزايدة إلى أدوات مؤتمتة قادرة على اكتشاف هذه الثغرات. تسهم هذه الأدوات في تعزيز أمان التطبيقات وتقليل الوقت والجهد الكبيرين اللذين يُستهلكان عادة في عمليات مراجعة الشفرة اليدوية، والتي غالبًا ما تكون مكلفة وتتطلب موارد عالية.

- يعرض هذا المستند إطار عمل يُدعى **NeuroAudit**، وهو نظام فعّال قائم على الشبكات العصبية، مصمم لتحليل شفرات PHPتلقائيًا واكتشاف الثغرات المحتملة من نوع SQLi و .XSSيقوم النموذج بتنفيذ تصنيف متعدد الفئات لتحديد ما إذا كانت قطعة الشفرة آمنة أو تحتوي على ثغرة، وإذا تبيّن أنها غير آمنة، يتم تحديد نوع الثغرة على وجه التحديد.

- تبدأ عملية الكشف بتحويل الشفرة إلى **شجرة البنية المجردة(AST)** ، التي تُعبر عن البنية النحوية والعناصر الدلالية للبرنامج. يتم بعد ذلك تحويل هذه التمثيلات إلى رموز (Tokenization) وتحليلها باستخدام نظام يعتمد على **الشبكات العصبية الالتفافية (CNN)** مع **شبكات الذاكرة طويلة وقصيرة الأمد ثنائية الاتجاه(BiLSTM)** ، وهو تكوين يسمح بالحفاظ على تسلسل الشفرة مع التقاط العلاقات بعيدة المدى بشكل فعّال.

- قمنا بتقييم أداء النظام باستخدام مجموعة اختبارات PHP – XSS و (2022) SQLi 1.0.0المقدمة من قاعدة بيانات مرجعية لضمان البرمجيات .(SARD) وقد أظهر NeuroAudit أداءً متميزًا، حيث حقق نسب دقة واسترجاع و F1 بلغت 82%، و98%، و90% على التوالي لاكتشافSQLi ، و100%، و97%، و99% لاكتشافXSS ، و96% عبر جميع المقاييس في حالة التصنيف المتعدد الفئات.

- كما تم التأكد من فعالية النموذج باستخدام أمثلة حقيقية مأخوذة من مستودعات GitHub وقاعدة بياناتExploit DB ، مما يعزز من موثوقية وكفاءة NeuroAudit في البيئات التطبيقية الواقعية.

جامعة حلوان
كلية الحاسبات والذكاء الاصطناعي
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NeuroAudit

AI Secure Code Reviewer

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